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ARE EFFICIENT AND FAIR:
IMPLICATIONS FOR TRANSFER
SCHEMES AND CARBON MARKETS**

Marc Fleurbaey, Ulrike Kornek and Ottmar Edenhofer

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Abstract

An international carbon pricing regime offers significant efficiency gains by avoiding climate change and reducing emissions at least cost. We clarify the role that country-specific prices play with respect to efficiency and burden sharing in an unequal world. Country-specific carbon prices are efficient and serve society's equity objectives if second-best constraints exclude optimal transfers to deal with pre-existing inequalities. This also holds true within the framework of carbon markets, where trade at a common price is just one allocation at the (constrained) Pareto-efficient frontier if the initial permit allocation is fixed. A common carbon price aligns with social objectives if transfers or initial permit allocations can be freely adjusted, and otherwise differentiated prices may be preferable. But the social welfare gains from differentiated carbon prices may vary depending on empirical facts. We present a theoretical model which connects the level of national carbon prices to the choice of social welfare functions that capture various equity principles. With a calibrated integrated assessment model we quantify country-specific carbon prices for a large set of countries and show that they can significantly promote social welfare.

JEL Classification: D62, D63, Q54, Q58

Keywords:

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When non-uniform carbon prices are efficient and fair: Implications for transfer schemes and carbon markets*

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Abstract

An international carbon pricing regime offers significant efficiency gains by avoiding climate change and reducing emissions at least cost. We clarify the role that country-specific prices play with respect to efficiency and burden sharing in an unequal world. Country-specific carbon prices are efficient and serve society's equity objectives if second-best constraints exclude optimal transfers to deal with pre-existing inequalities. This also holds true within the framework of carbon markets, where trade at a common price is just one allocation at the (constrained) Pareto-efficient frontier if the initial permit allocation is fixed. A common carbon price aligns with social objectives if transfers or initial permit allocations can be freely adjusted, and otherwise differentiated prices may be preferable. But the social welfare gains from differentiated carbon prices may vary depending on empirical facts. We present a theoretical model which connects the level of national carbon prices to the choice of social welfare functions that capture various equity principles. With a calibrated integrated assessment model we quantify country-specific carbon prices for a large set of countries and show that they can significantly promote social welfare.

Keywords: carbon price, burden sharing, fairness.

JEL: D62, D63, Q54, Q58.

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1 Introduction and relation to literature

1.1 Introduction

The question of whether carbon prices should be equal in all parts of the world has been debated among economists for several decades already. The seminal contribution of Chichilnisky and Heal (1994) has sparked interest in differential prices aligned to social priorities on the one hand. On the other hand, the efficiency argument that total abatement cost is minimized only when marginal cost is equalized everywhere has motivated many advocates of a uniform price (e.g., Cramton and Stoft 2015; Blanchard et al. 2022). This issue has strong ramifications for the geopolitics of climate change mitigation. A uniform carbon price could impose disproportionate abatement costs on the developing world, relative to current income, and the transfers that would be needed to compensate for such imbalance are unrealistically high. As argued in Stiglitz (2019) and Bauer et al. (2020), allowing for moderate deviations from a uniform carbon price would bring down the needed transfers considerably, without generating staggering inefficiency.

With this paper, we revisit the theoretical underpinnings and practical implications of this issue and make three contributions.

First, we clarify the distinct parts that efficiency and equity considerations play in this topic. In particular, efficiency depends on the prevailing constraints and it is useful to distinguish various constrained settings, and thus various Pareto frontiers, in which the design of carbon pricing may take place. A key parameter in the debate is whether transfers are possible or not. Equity cannot be separated from efficiency if transfers are limited. What is generally missing from the literature is recognition of the full extent to which markets for emission permits are vehicles for particular payment transfers that may supplement, or substitute for, direct transfers.

Second, we introduce the possibility of having an emission trading system with differential prices. While this possibility has been mentioned in the literature (in particular in Chichilnisky et al. 2000), under the label of Lindahl pricing, we believe that the potential scope and impact for such mechanisms has been under-appreciated. When there are strong constraints on direct transfers and on the distribution of permit endowments, the introduction of differential prices provides ways to promote social welfare beyond what a uniform-price market can achieve.

Third, equity considerations in typical theoretical analyses of carbon pricing are merely embodied in social weights in an additively separable formulation (or approximation) of social welfare, and such weights are generally hard to connect to concrete issues about inequalities along various dimensions of living standards, historical responsibilities, needs and capacities, and so on. In this paper we review the main equity considerations that pertain to the liter-

ature on burden-sharing, and provide the missing connection to the design of social welfare functions that capture various equity principles.

We quantify our theoretical results with a calibrated integrated assessment model. In the model, optimal carbon prices and transfers, if available, implement an equity principle and are (constrained) Pareto-efficient. We find that low-income countries tend to have a lower burden compared to high-income countries to achieve a range of equity principles. A lower burden arises either from a lower carbon price or a higher transfer received, or both. In the case in which transfers (or a permit market) are not available, social welfare improves only moderately when country-specific carbon prices achieve equity, compared to a globally uniform price. Welfare gains are moderate because either the uniform-price allocation is close to the considered equity principle (for example demanding equal abatement costs as a share of income) or the equity principle demands large inequality in country-specific carbon prices, resulting in large losses from higher total abatement costs (for example achieving equal emissions with a range of national prices between 1 and 15000 USD/tCO₂). With a permit market in place, we find that country-specific carbon prices can significantly increase social welfare, by up to 1000 USD per person. Welfare gains can be large because transfers through permit trade are available in addition to country-specific prices to address equity. Lastly, if transfers are freely available, we find that social welfare can improve by more than 5000 USD per person. As one can fully separate equity from efficiency in this case, large gains in social welfare are possible. However the necessary transfers can be unrealistically high (up to 600% of income received).

Our results show that recommendations for harmonizing internationally differential carbon prices should be accompanied with recommended transfer schemes. Empirically, carbon prices tend to be higher in richer countries with better governance systems (World Bank, 2023; Levi et al., 2020), a relationship for which we provide ethical support if transfers are limited. If carbon prices were harmonized, lower-income countries would need to be compensated via transfers from higher-income countries. Such transfers are already in place, for example within the European Union through its Just Transition Fund and Social Climate Fund or to a limited extent through the Paris Agreement’s Green Climate Fund. Our results show that the scale of transfers needed to reach a fair allocation under a common carbon price may be too large and country-specific prices may be politically more feasible to implement equity between countries in climate change mitigation.

Our results also show the role that differential prices can play for designing international carbon markets. Without differential prices, for example, joining permit trading schemes between multiple countries would fix national contributions – both financially and environmentally – through the common permit price and given initial permit endowments. However, some countries may wish to increase their contributions after observing that the market out-

come means too little domestic action. This can be done by increasing domestic (shadow) prices, retaining a common permit price. For example, Germany’s early coal phase-out alters the distribution of emission reduction efforts among countries in the European Union’s Emissions Trading Scheme (Anke et al., 2020), just as the UK carbon price floor (Hirst and Keep, 2018; Leroutier, 2022). Our results show that trade at differential prices can further promote equity while retaining efficiency.

The paper is structured as follows. After relating our analysis to previous literature in the next subsection, Section 2 contains a brief introduction of the theoretical model. In section 3, we propose an analysis of efficiency and equity in general terms, for different cases involving possible constraints on transfers, permit allowances, and carbon prices. In section 4, we formalize relevant principles of equitable burden-sharing and show how they inform the social weights that determine socially optimal allocations in the theory developed in section 3. In section 5, a numerical simulation illustrates the level and distribution of carbon prices that would follow from various equity principles, and show how market transactions could lead to substantial transfers of resources across countries. Section 6 offers conclusions and perspectives for future research.

1.2 Relation to previous literature

Here is a brief description of how this paper relates to the literature. In the framework initiated by Chichilnisky and Heal (Chichilnisky and Heal 1994, 2000, Chichilnisky et al. 2000, Sheeran 2006, Sandmo 2007), two cases are contrasted. When transfers are possible and costless, efficiency requires equalizing marginal abatement costs and social optimality requires equalizing the marginal social value of money, across all countries. When transfers are impossible, efficiency is compatible with different marginal costs and social optimality requires them to be inversely proportional to marginal social values of money. d’Autume et al. (2016) retain this taxonomy in a paper considering households within countries using distortionary taxes for within-country redistribution, as well as local effects of externalities (e.g., air pollution). They show that if international transfers are costless, the local tax on the externality has a component that represents the global externality and is the same across countries. In contrast, if transfers are not possible, then differential taxes, even for the global impact, are generally warranted. Bataille et al. (2018) rely on the same theoretical framework and examine how to combine pricing mechanisms with other instruments in policy packages, without discussing differential prices. In this paper, we propose to consider additional cases, in particular involving markets with constraints on prices or on endowments. Some additional cases have appeared in the literature. Heal (2000) introduces a case in which the total quantity of abatement is fixed, but then focuses again on the distinction between costless and impossible transfers, replicating the conclusions of earlier papers. Closer to our focus, Shiell (2003)

introduces a case in which a uniform-price market prevails, but direct transfers are not possible and permit endowments are constrained to be non-negative, so that only a subset of the first-best frontier is reachable.

In the framework offered by this particular literature, efficiency is often depicted by the general condition that marginal costs should be inversely proportional to marginal social values of money, and the first-best situation is viewed as the special case in which the latter are equalized thanks to transfers and therefore the former are equal as well. We propose a revision to this framing. The first-best situation is not a special case of a general inverse proportion rule, because there are two different Pareto frontiers, one for the costless-transfer case, and one for the no-transfer case. That they intersect at one point does not make the first best a special case of the no-transfer case. Moreover, the inverse proportion rule is ill-suited to qualify as an efficiency criterion, because marginal social values of money are relevant to social optimality, not to efficiency. That every efficient allocation is optimal for some social welfare function does not require describing efficiency in terms of the implicit social weights, and insofar as this is possible, it is better to provide a direct description of the characteristics of efficient allocations. In this paper, we provide a direct characterization of efficiency in terms of intrinsic properties of the allocations. And in relevant cases, we will find that no inverse proportion rule holds at efficient allocations.

The identification of efficiency with the inverse proportion rule leads Chichilnisky et al. (2000) and Heal and Lin (2000) to describe emission markets with differential prices (Lindahl pricing) as making it possible to obtain efficiency gains through restoring the inverse proportion rule. In this paper, we show that differential prices, compared to a uniform-price allocation, make it possible to pursue redistributive purposes, not efficiency gains. As a matter of fact, compared with the no-transfer allocations (in which the inverse proportion rule holds), opening a market (with uniform or differential prices) does generate Pareto-superior allocations. In other words, the relevant Pareto frontier for the analysis of market allocations is above, not on or below, the Pareto frontier for the no-transfer case, which is very intuitive since, starting from a no-transfer efficient allocation, and taking current emissions as the initial allowance, opening a market offers new opportunities for all traders and will generally make each of them better off. In fact, market trade offers a limited but significant way to implement transfers between agents, and thus performs a distributive as well as an efficiency function (a fact that was noted as early as in Smith's *Theory of Moral Sentiments*). Therefore it is understandable that the Pareto frontier for market allocations is closer to the costless-transfer frontier than the no-transfer frontier.

The equity principles considered in this paper echo those found in the burden-sharing literature (e.g. Bode 2004; Lange et al. 2007; Mattoo and Subramanian 2010; Bretschger 2013), and bearing on the distribution of effort, emissions, historical responsibility, and final con-

sumption. The literature generally focuses on finding precise distributive patterns (equality, proportionality), although Bretschger (2013) proposes to use a mixed index combining all the principles to compute the degree of fairness exhibited by any allocation. We find that in important cases, the proposed patterns conflict with efficiency or even with feasibility, and we propose a way around this difficulty by relying on the equivalence approach that is popular in the theory of fair allocation (Moulin 1987, Maniquet and Sprumont 2004). This approach consists in using the fairness principle to define reference situations to which the actual allocation is compared on the basis of the individual countries' utility, and policy analysis then seeks to find an allocation that is indifferent, for each country, to an allocation that satisfies the target fairness principle.

2 Model

We model a global public good, provided through contributions by a set of countries. The countries $i = 1, \dots, N$ have utility $n_i U_i(c_i, X)$ over their private per capita consumption c_i and a public good X (n_i denotes the population size). We ignore inequalities within countries in this paper. The public good (e.g., abatement of emissions) is made of their individual contributions $X = \sum_i n_i x_i$, where $x_i \geq 0$ denotes per capita contributions.¹ Their abatement costs may differ, and absent any transfers or emissions trading, their consumption is equal to their given endowment e_i diminished of the abatement cost $g_i(x_i)$: $c_i = e_i - g_i(x_i)$. We assume throughout this paper that, for all i , U_i is continuously differentiable and strictly quasi-concave, while g_i is continuously differentiable, convex, and such that $g_i(0) = 0$. In Sections 2 through 4, all individual country variables (consumption, abatement, utility) are defined per capita.

Different endowments (e_i) capture pre-existing inequalities between countries in our model. Different national costs of abatement (g_i) and benefits derived from the global public good (U_i) capture inequalities that relate to the problem of climate change mitigation. We will be more specific about different sources of inequalities that are particularly important in the climate context in Section 4.

Pareto efficiency is defined as usual (no agent can be made better off without making another agent worse off). Social optimality is defined in terms of maximizing a social welfare function. We assume that the social welfare function is of the weighted-sum form

$$SWF = \sum_i \alpha_i n_i U_i(c_i, X), \quad (1)$$

¹The conditions throughout the paper assume that abatement is positive unless stated otherwise.

where the weights α_i are normalized: $\sum_i \alpha_i n_i = n$, $n = \sum n_i$. For social optimality, the weights are parameters of the problem and the results will depend on them. The welfare weights represent social preferences for distributional justice among countries. We briefly revisit the case of other social welfare functions in Section 3.

To analyze Pareto-efficiency and social optimality, we distinguish three cases. In the first, there are no international transfers, and countries need to rely on their initial endowments for consumption. In the second, costless international transfers are possible such that any allocation only needs to fulfill the aggregate constraint $\sum_i n_i t_i = 0$, where t_i is the per capita transfer received by country i . In the third, countries have, additional to their initial endowment e_i , an arbitrarily given initial allocation of permits x_i^* in an international carbon market.² By assumption, no direct transfers or redistribution of permits are possible in this case, but trade in permits induces limited international transfers between countries, via payments. And for this third case, we will compare the subcase in which differential prices (p_i) are allowed with the subcase in which a uniform price (p) applies. The market equilibrium requires that total actual abatement be equal to the total allowance $\sum_i n_i (x_i - x_i^*) = 0$, but also, when differential prices are used, that the sum of induced payments adds up to zero $\sum_i p_i n_i (x_i - x_i^*) = 0$ (this condition is automatically satisfied under uniform pricing when the market is at equilibrium).

In the remainder of this paper, the three cases are respectively labelled *no transfers*, *unlimited transfers*, and *market-constrained transfers*. To recap, the budget of country i and additional resource constraints at the economy level are listed in Table 1 for the three cases.

Case	Country budget	Economy-wide constraints
No transfers	$c_i = e_i - g_i(x_i)$	
Unlimited transfers	$c_i = e_i + t_i - g_i(x_i)$	$\sum_i n_i t_i = 0$
Market-constrained transfers	$c_i = e_i + p_i(x_i - x_i^*) - g_i(x_i)$	$\sum_i p_i n_i (x_i - x_i^*) = 0$ $\sum_i n_i (x_i - x_i^*) = 0$

Table 1: Budget and other constraints

3 Pareto-efficient and socially optimal allocations

This section discusses the relationship between Pareto-efficient and socially optimal allocations in all three cases studied in this paper. We first briefly discuss the no-transfer and unlimited-transfer cases in section 3.1 which are familiar in the literature. The third case of limited transfers in a carbon market is discussed in section 3.2.

²These are abatement permits, which are not as common as emission permits, but this is obviously equivalent, and adopted here to save notations in the first part of the paper.

3.1 No international transfers vs. unlimited transfers

To find all Pareto efficient allocations, one maximizes the utility of each individual country under the constraint of keeping the other countries' utilities at given levels.³ Table 2 summarizes the key elements of efficiency analysis, using the notations $U_{ic} = \frac{\partial}{\partial c_i} U_i(c_i, X)$ and $U_{jX} = \frac{\partial}{\partial X} U_j(c_j, X)$ for the first-order conditions (FOC).

	No transfers	Unlimited transfers
Objective	$\text{Max}_x U_1(c_1, X)$	$\text{Max}_{t,x} U_1(c_1, X)$
Constraints (multipliers)	$U_i(c_i, X) \geq u_i, \quad \forall i \neq 1 \ (\beta_i)$ $c_i = e_i - g_i(x_i), \quad \forall i$ $X = \sum_i n_i x_i$	$U_i(c_i, X) \geq u_i, \quad \forall i \neq 1 \ (\beta_i)$ $c_i = e_i + t_i - g_i(x_i), \quad \forall i$ $X = \sum_i n_i x_i$ $\sum_i n_i t_i = 0 \quad (\lambda)$
FOC ($\beta_1 = 1$)	$\beta_i U_{ic} g'_i \frac{1}{n_i} = \sum_j \beta_j U_{jX}, \quad \forall i$	$\beta_i U_{ic} g'_i \frac{1}{n_i} = \sum_j \beta_j U_{jX}, \quad \forall i$ $\beta_i U_{ic} \frac{1}{n_i} = \lambda, \quad \forall i$

Table 2: Pareto-efficiency, under constraints of no and unlimited transfers.

In both cases, the first FOC implies a variant of the Bowen-Lindahl-Samuelson (BLS) condition – by rearranging and summing over the first FOC of each country –, which is therefore a necessary condition of Pareto-efficiency:

$$\sum_{i=1}^N \frac{n_i U_{iX}}{U_{ic} g'_i(x_i)} = 1.$$

Under the convexity and smoothness assumptions made in this paper, this is also a sufficient condition in case of no transfers. Marginal costs of abatement $g'_i(x_i)$ of each country directly enter the BLS condition. This allows us to draw immediate conclusions about how carbon prices differ by country, because prices are equal to the marginal costs of abatement when agents choose their abatement level under a budget constraint (as later formalized in section 3.2). While there is an allocation in which marginal costs are equal, most allocations satisfying the BLS condition do not have equal marginal costs. To see this clearly, take the specific example of $U_i(c_i, X) = U_i(e_i - g_i(x_i) + b_i(X))$, where $b_i(X)$ is the monetary equivalent of the benefit from the public good. The BLS condition, in this example, reads $\sum_{i=1}^N \frac{n_i b'_i(X)}{g'_i(x_i)} = 1$, which is fulfilled, among a continuous set of other allocations, with a uniform carbon price $g'_i(x_i) = \sum_{j=1}^N n_j b'_j(X)$, as well as with differentiated carbon prices $g'_i(x_i) = N n_i b'_i(X)$.

For the case with unlimited transfers, Pareto-efficient allocations additionally exhibit equal marginal abatement costs $g'_i(x_i) = g'_j(x_j) \ \forall i, j$, which implies that carbon prices are equal

³An alternative method consists in maximizing the social welfare function from Eq. (1), for arbitrary weights. There is an obvious relationship between such weights and the Lagrange multipliers in the program laid out in the main text.

across countries. This comes from the second FOC, which, inserted into the first one, transforms it into $g'_i \lambda = \sum_j \beta_j U_{jX}$. In this case, thanks to our assumptions again, the marginal-cost equality condition and the BLS condition jointly make a sufficient condition for efficiency.

As can be seen from this analysis, Pareto efficiency recommends equal carbon prices only in the first-best case with costless transfers, whereas no conclusion about the distribution of carbon prices can be drawn from Pareto-efficiency in the absence of international transfers. Constrained Pareto-efficiency in case of no transfers allows equal marginal costs, but most allocations on the constrained Pareto-frontier do not have equal prices, and (constrained) Pareto efficiency is compatible with very different distributions of carbon prices. To answer the question whether carbon prices should be equalized, or follow a specific distribution pattern, we need to rely on social optimality analysis. All socially optimal allocations are Pareto efficient in either case. Preferences over distributional justice, through the choice of welfare weights, pick which Pareto efficient allocation is also socially optimal.

When maximizing the social welfare function in Eq. 1, both in the no-transfer and unlimited-transfer cases, the first-order condition with respect to abatement is, for all i , similar as in Table 2 but now with exogenously given weights α_i :

$$\alpha_i U_{ic} g'_i(x_i) = \sum_{j=1}^N \alpha_j n_j U_{jX}, \quad (2)$$

The equation reads as

$$MSV_i g'_i(x_i) = MSV_X,$$

where $MSV_i = \alpha_i U_{ic}$ is the marginal social value of an individual's consumption in i , and $MSV_X = \sum_j \alpha_j n_j U_{jX}$ is the marginal social value of the public good. The underlying programs are displayed in Table 3.

	No transfers	Unlimited transfers
Objective	$\text{Max}_x \sum_i \alpha_i n_i U_i(c_i, X)$	$\text{Max}_{t,x} \sum_i \alpha_i n_i U_i(c_i, X)$
Constraints (multipliers)	$c_i = e_i - g_i(x_i), \quad \forall i$ $X = \sum_i n_i x_i$	$c_i = e_i + t_i - g_i(x_i), \quad \forall i$ $X = \sum_i n_i x_i$ $\sum_i^n n_i t_i = 0 \quad (\lambda)$
FOC	$\alpha_i U_{ic} g'_i = \sum_j \alpha_j n_j U_{jX}, \quad \forall i$	$\alpha_i U_{ic} g'_i = \sum_j \alpha_j n_j U_{jX}, \quad \forall i$ $\alpha_i U_{ic} = \lambda, \quad \forall i$

Table 3: Social optimality without and with unlimited transfers cases.

If transfers are cost-less, the model with unlimited transfers additionally requires the equalization of marginal social values of consumption: $MSV_i = MSV_j \quad \forall i, j$. This is immediately stated in the second FOC: $\alpha_i U_{ic} = \lambda$. As above for Pareto-efficiency, marginal costs of abatement are therefore equalized. Carbon prices are equal across countries, and this requires

specific transfers leading to equal MSV_i .

In the model without transfers, the equation $g'_i(x_i) = \frac{MSV_X}{MSV_i}$ derived from (2) characterizes the social optimum. One thus finds that marginal costs must be inversely proportional to the MSV_i .⁴ If, for instance, welfare weights reflect equality ($\alpha_i = 1$), poorer countries tend to have a lower marginal cost of abatement as their marginal utility of consumption tends to be higher. The optimal allocation implements an implicit redistribution from rich to poor countries through the allocation of abatement. This argument remains valid for a much more general class of social objectives of the type $\sum_i \frac{n_i}{n} \varphi(U_i(c_i, X))$, $n = \sum_j n_j$, where φ is a concave transform embodying aversion to inequality in utility. This type of “prioritarian” social welfare function is increasingly popular in welfare analysis, including in climate economics (e.g., Adler and Treich 2015; Adler et al. 2017; Budolfson et al. 2017).

Social optimality does recommend equal prices even without transfers if the prevailing level of inequality among countries is considered optimal. Formally, this can be implemented by adopting Negishi weights α_i that are inversely proportional to U_{ic} , inducing equal MSV_i and therefore equal marginal costs for all countries.

Our conclusions about Pareto efficiency and social optimality are gathered in Table 4. Specific conclusions about the distribution of carbon prices (and marginal abatement costs) are obtained in the unlimited-transfer case (as a consequence of efficiency considerations) and under social optimality in the no-transfer case, but not in the Pareto efficiency analysis for the latter case.

	No transfers	Unlimited transfers
Pareto efficiency	BLS condition	BLS condition equal g'_i
Social optimality	BLS condition g'_i inv. prop. to MSV_i	BLS condition equal g'_i equal MSV_i

Table 4: Characterizing efficiency and social optimality

We conclude this section by providing a simple analytical example illustrating these points.

Example Let $n_i = 1$ for all countries, and assume that $U_i(c_i, X) = \ln c_i + \theta_i X$, while $g_i(x_i) = \frac{e^{\gamma_i x_i} - 1}{\gamma_i}$, with $\gamma_i > 0$, and $X = \sum_i x_i$. Marginal costs are $e^{\gamma_i x_i}$, an expression which converges toward the constant-marginal-cost function $g_i(x_i) = x_i$ when $\gamma_i \rightarrow 0$.

⁴In the literature, this inverse proportionality rule is often presented as an efficiency rule, but we suggest it is less confusing to view it as a social optimality rule, because it is only in the context of social optimality that weights are determined. The fact that every efficient allocation is socially optimal for some weights is true, but since those weights are endogenous to the allocation, it makes little sense to refer to them to determine the allocation itself. The BLS condition is the true embodiment of efficiency in the absence of transfers.

One can interpret γ_i as the percentage of increase of the marginal cost with x_i . We assume $e_i > 1$ and $\theta_i \geq 1$ for all i .

The BLS condition, in this example, reads $\sum_i \frac{\theta_i c_i}{e^{\gamma_i x_i}} = 1$, which after inserting $c_i = e_i + t_i - \frac{e^{\gamma_i x_i} - 1}{\gamma_i}$, becomes

$$\sum_i \frac{\theta_i \left(e_i + t_i + \frac{1}{\gamma_i} \right)}{e^{\gamma_i x_i}} = 1 + \sum_i \frac{\theta_i}{\gamma_i}.$$

This shows that a sort of harmonic mean of marginal cost must satisfy a given value, leaving it possible, quite clearly, to have unequal marginal costs.

In the model with unlimited transfers, the $e^{\gamma_i x_i}$ are equal across individuals and one obtains a precise solution for the x_i , for any given distribution of transfers:

$$x_i = \frac{1}{\gamma_i} \ln \frac{\sum_j \theta_j \left(e_j + t_j + \frac{1}{\gamma_j} \right)}{1 + \sum_j \frac{\theta_j}{\gamma_j}}.$$

Observe how the optimal quantity of public good, in this example, depends on the correlation between endowments (e_i) and preferences (θ_i).

In the social optimality approach with α_i given, for the case without transfers one obtains

$$x_i = \frac{1}{\gamma_i} \ln \frac{\theta_i \left(e_i + \frac{1}{\gamma_i} \right)}{\alpha_i + \frac{\theta_i}{\gamma_i}} \text{ and } MSV_i = \frac{\alpha_i + \frac{\theta_i}{\gamma_i}}{e_i + \frac{1}{\gamma_i}}.$$

As is intuitive, the agent i 's MSV_i is increasing in α_i and in θ_i , and decreasing in e_i , and the opposite is true for the contribution x_i .

In case of unlimited transfers, the MSV_i are equalized, and thus one obtains, in addition to the efficient values for x_i provided earlier:

$$t_i = \left(\frac{\alpha_i + \frac{\theta_i}{\gamma_i}}{MSV_i} \right) - \frac{1}{\gamma_i} - e_i \text{ and } MSV_i = \frac{1 + \sum_j \frac{\theta_j}{\gamma_j}}{\sum_j e_j + \sum_j \frac{1}{\gamma_j}}.$$

Let us now examine how the solution moves when $\gamma_i \rightarrow 0$, so that differences in marginal costs vanish. Without transfers one obtains the socially optimal allocation

$$x_i = e_i - \frac{\alpha_i}{\sum_j \theta_j \alpha_j}, \quad MSV_i = \sum_j \theta_j \alpha_j.$$

There is no difference between no and unlimited transfers because the MSV_i are already equal when transfers are absent. In this particular case, there is no need to redistribute

endowments because abatement efforts can be used to perform the same redistribution.

3.2 Market-constrained international transfers

One key argument in the literature and public debate in favor of a uniform price is that linking markets and allowing for trade of emission permits across regions starting with different carbon prices will create new opportunities, possibly even a Pareto improvement.⁵ In this section we show that even with a permit market in place, differentiated carbon prices are generally Pareto-efficient and socially superior to a uniform price if initial allowances are fixed. In this respect, a permit market with fixed allowances is similar to the case without transfers of the previous section, but it generates allocations on a superior Pareto frontier. In contrast, a uniform carbon price is, generally, socially optimal and Pareto-efficient only if initial allowances can be freely adjusted, which creates a situation that is equivalent to the availability of costless international transfers. The permit market will then be used to implement the social optimum, with the initial allowance used to deal with pre-existing inequalities.

Let us introduce a permit market in the current model and examine the Pareto-efficient allocations. Let $x^* = (x_i^*)_{i=1,\dots,N}$ denote the initial allowances given to the countries and open a market for emission permits. This gives i the new budget $c_i = e_i - g_i(x_i) + p_i(x_i - x_i^*)$, where p_i is the price of permits, which may or may not be specific to i . Assuming that countries don't take account of their small impact on the total contribution to the public good X , they will thus choose x_i so that

$$g'_i(x_i) = p_i.$$

Since x_i^* lies among the options in the budget, every i is at least as well off as with x_i^* . For instance, if x^* corresponds to an efficient allocation of abatement efforts in the case without transfers, the market reaches a higher Pareto frontier.

Let us distinguish three cases, depending on how much flexibility one has about changing X and x^* :

1. Unconstrained: There is no restriction on X and x^* . This case is equivalent to the case with unlimited transfers of the previous section, with allowances serving to implement transfers across countries. There is no use for differentiated prices in this case.

⁵There is another argument in favor of a single price, which relies on the assumption that in a second-best setting, the government does not observe MSV_i but instead a proxy such as income. In this context, the best redistribution tool is the income tax and differential prices are useless as an additional redistributive instrument (Gauthier and Laroque 2009; Jacobs and De Mooij 2015; Fleurbaey and Kornek 2021). This argument has a limited scope, however, since it relies on the key assumption that the income tax can be used as flexibly as needed. In the international context, such a global redistribution mechanism does not exist, and even at the national level, political constraints generally impede a fine-tuning of the income tax.

2. Fixed X : The total quantity $\sum_i n_i x_i^*$ is fixed, but the distribution x^* can be adjusted freely under this constraint. Again, there is no use for differentiated prices in this case.⁶ Given that X is fixed and marginal costs $g'_i(x_i)$ are equalized, the allocation of effort x is always the same, independent of allowances x^* . Thus, all allocations in this case can also be produced by fixing X, x and implementing direct transfers among countries.
3. Fixed x^* (and X): This is the market-constrained case introduced in section 2. If a single price prevails, typically only one allocation is obtained. In this case, allowing for differentiated prices generates a greater set of possible allocations. We focus on this case in the rest of this section.

It must be emphasized that the Pareto-frontier of case 3 (market-constrained transfers) always lies below that of case 2 (fixed X but unlimited transfers). Consider an allocation x_0 with differential prices. Let $x^* = x_0$, and implement the transfers $t_i = p_i n_i (x_i - x_i^*)$. In absence of a market, this allocation is identical to the allocation x_0 with differential prices. Then, open a market with a single price. After the market is open, the allocation of effort is back at the distribution x that prevails in all efficient allocations of case 2, since X always remains the same in all these allocations. Now, every country is at least as well-off as at the initial allocation before the market opened. But the new allocation belongs to the frontier of case 2, since it has the right distribution of effort, associated with suitable transfers (which can be performed via a suitable choice of initial permits). This proves that, even in the constrained case in which X is given, it is always better to perform transfers (directly or via the distribution of permits) and maintain a single price, than relying on multiple prices. Our interest in multiple prices comes from the possibility that such transfers may be politically impossible, in contrast with price adjustments.

Let us seek the Pareto-efficient allocations in the market-constrained case, under the balancing constraints that $\sum_i n_i x_i = \sum_i n_i x_i^*$ and $\sum_i p_i n_i (x_i - x_i^*) = 0$. Observe that not all differentiated-price equilibria are efficient under these constraints. For instance, the initial allocation x^* is an equilibrium for prices that keep everyone from trading, but this is generally dominated by the uniform-price equilibrium.

We keep assuming that countries are price takers, so that for all i , $g'_i(x_i) = p_i$. Let $x_i(p_i) = (g'_i)^{-1}(p_i)$. This function is well-defined and increasing when cost functions are strictly convex,

⁶Indeed, consider a market equilibrium with differentiated prices. At this equilibrium, one has $\sum_i n_i x_i = \sum_i n_i x_i^*$. Now take the distribution x as the new allowance, which is possible under the fixed X constraint, and allow the agents to trade again, at a uniform price. They can only end up in a Pareto superior allocation.

which we assume here. The Lagrangian is

$$L = \sum_i \beta_i U_i \left(e_i - g_i(x_i(p_i)) + p_i n_i (x_i(p_i) - x_i^*), \sum_i n_i x_i^* \right) + \mu \left(\sum_i n_i x_i(p_i) - \sum_i n_i x_i^* \right) - \lambda \sum_i p_i n_i (x_i(p_i) - x_i^*)$$

The first-order condition with respect to p_i reads:

$$\left(\beta_i U_{ic} \frac{1}{n_i} - \lambda \right) (x_i(p_i) - x_i^*) + (\mu - \lambda p_i) x_i'(p_i) = 0. \quad (3)$$

	Market-constrained transfers
Objective	$\text{Max}_p U_1(c_1, X)$
Constraints (multipliers)	$U_i(c_i, X) \geq u_i, \quad \forall i \neq 1 (\beta_i)$
	$c_i = e_i - g_i(x_i) + p_i(x_i - x_i^*), \quad \forall i$
	$X = \sum_i n_i x_i$
	$g_i'(x_i) = p_i \quad \forall i$
	$\sum_i p_i n_i (x_i - x_i^*) = 0 \quad (\lambda)$
FOC ($\beta_1 = 1$)	$\sum_i n_i (x_i - x_i^*) = 0 \quad (\mu)$
	$\left(\beta_i U_{ic} \frac{1}{n_i} - \lambda \right) (x_i(p_i) - x_i^*) + (\mu - \lambda p_i) x_i'(p_i) = 0, \quad \forall i$

Table 5: Pareto-efficiency, market-constrained case.

We claim that this FOC is reached at every equilibrium if and only if it satisfies the following condition:

$$\forall i, j, x_i(p_i) - x_i^* \geq 0 \geq x_j(p_j) - x_j^* \Rightarrow p_j - p_i \leq \frac{x_i(p_i) - x_i^*}{x_i'(p_i)} - \frac{x_j(p_j) - x_j^*}{x_j'(p_j)}. \quad (4)$$

This condition controls the price difference between j and i when i and j are on opposite sides of the permit market. As a special case, it implies that all agents who do not trade at the equilibrium should face the same price, and thus, in particular, that the equilibrium which keeps everyone at the initial x_i^* with different prices is not efficient, as already explained.

Checking that (3) implies (4) is done by deriving from the former the equality

$$p_j - p_i = \frac{x_i(p_i) - x_i^*}{x_i'(p_i)} - \frac{x_j(p_j) - x_j^*}{x_j'(p_j)} + \beta_j \frac{U_{jc}}{\lambda n_j} \frac{x_j(p_j) - x_j^*}{x_j'(p_j)} - \beta_i \frac{U_{ic}}{\lambda n_i} \frac{x_i(p_i) - x_i^*}{x_i'(p_i)}$$

where one checks that when $x_i(p_i) - x_i^* \geq 0 \geq x_j(p_j) - x_j^*$, the last two terms are negative, thus implying the inequality in (4).

Let us also show why condition (4) implies the satisfaction of FOC (3). First observe that

since Eq. 3 is homogeneous in (β, μ, λ) , one can let $\lambda = 1$ and only seek (β, μ) . Now, (4) implies that one can pick μ such that:

$$\max \left\{ p_i + \frac{x_i(p_i) - x_i^*}{x_i'(p_i)} | x_i(p_i) - x_i^* \leq 0 \right\} \leq \mu \leq \min \left\{ p_i + \frac{x_i(p_i) - x_i^*}{x_i'(p_i)} | x_i(p_i) - x_i^* \geq 0 \right\}$$

By this choice of μ , one has, whenever $x_i(p_i) - x_i^* \neq 0$:

$$\forall i, \frac{(p_i - \mu) x_i'(p_i)}{x_i(p_i) - x_i^*} + 1 \geq 0$$

so that one can find $\beta_i \geq 0$ such that

$$\beta_i \frac{U_{ic}}{n_i} = \frac{(p_i - \mu) x_i'(p_i)}{x_i(p_i) - x_i^*} + 1,$$

which is equivalent to Eq. 3. When $x_i(p_i) - x_i^* = 0$, then $p_i = \mu$, and Eq. 3 is again satisfied.

As the above condition clearly shows, prices are generally differentiated at the Pareto-efficient frontier, and the uniform-price equilibrium is, generally, just one particular point on this frontier.

Turning to socially optimal allocations, for the social welfare function $SWF = \sum_i \alpha_i n_i U_i$, the conditions characterizing social optimality can be found by substituting $\beta_i U_{ic} \frac{1}{n_i}$ for $\alpha_i U_{ic} = MSV_i$ in Eq. 3, where the welfare weights α_i are given. We thus have, at the socially optimal allocation,

$$p_i = \frac{\mu}{\lambda} + \frac{(MSV_i - \lambda)(x_i(p_i) - x_i^*)}{\lambda x_i'(p_i)},$$

which, after noting that μ is the marginal social value of X , while λ measures the marginal value of money for social welfare, and thus averages the MSV_i over the countries, can be written as

$$p_i = \frac{MSV_X}{\overline{MSV}} + \frac{\left(\frac{MSV_i}{\overline{MSV}} - 1 \right) (x_i(p_i) - x_i^*)}{x_i'(p_i)} \quad (5)$$

where $\overline{MSV} = \sum_i \frac{n_i}{n} MSV_i$.

This formula can be contrasted with the socially optimal allocations without transfers (i.e., in the absence of the carbon market). The term $\frac{MSV_X}{\overline{MSV}}$ is the counterpart of the ratio $\frac{MSV_X}{MSV_i}$ from the case without transfers, but with an averaged denominator. This kills the inverse proportion rule linking differential carbon prices to marginal social value. The latter actually appears in the second term of the formula, which is new and takes account of the transfer value of having $x_i - x_i^* > 0$. This value is nil if MSV_i has the average value,⁷ is positive if MSV_i

⁷Note that, here as in the previous sections, under Negishi weights which equalize the MSV_i across agents, the prices would be equal.

has greater-than-average value, and conversely. Carbon prices should, therefore, generally be differentiated by country, as a way to partly alleviate the constraints bearing on transfers and allowance distribution.

Observe that the differentiated part of equation (5) can be approximated by the expression

$$\left(\frac{MSV_i}{\overline{MSV}} - 1\right) (g'_i(x_i) - g'_i(x_i^*))$$

since $1/x'_i(p_i)$ is equal to $g''_i(x_i)$. This gives another intuitive interpretation to the pricing formula: The differentiated part of the price should reflect the social priority of the country (first bracket) and the variation in marginal abatement cost it incurs from its initial permit endowment (second bracket). A country that abates more than its initial endowment and has greater social priority should benefit from a proportionally higher price when it sells emission permits.

Our conclusions are summarized in Table 6.

	Market-constrained transfers
Pareto efficiency	$x_i - x_i^* \geq 0 \geq x_j - x_j^* \Rightarrow p_j - p_i \leq \frac{x_i - x_i^*}{x'_i} - \frac{x_j - x_j^*}{x'_j}$.
Social optimality	$p_i = \frac{MSV_X}{\overline{MSV}} + \frac{(\frac{MSV_i}{\overline{MSV}} - 1)(x_i - x_i^*)}{x'_i}$.

Table 6: Characterizing efficiency and social optimality

The results of this section can be illustrated with the model from the example in section 3.1. Unfortunately, illustrating differential prices graphically is a little difficult, because with two agents, it is impossible to have a multiple-price permit market.⁸ Let us therefore consider a three-agent case, with $\gamma = (1, 2, 2)$, $\theta = (2, 2, 1)$ and endowments $(20, 40, 40)$. We start from a constrained optimal allocation in which $x = (2.86, 2.07, 1.96)$, which is socially optimal in the constrained case for welfare weights $\alpha = (0.4, 0.3, 0.3)$, and consider opening a permit market (keeping the initial abatement efforts as the permit allowances). One can use x_1 as the exogenous variable to describe all possible equilibria, since $x_3 = X - x_1 - x_2$, and the remaining equation $\sum_i p_i (x_i - x_i^*) = 0$ then determines x_2 . As it turns out, there are two roots for x_2 , and therefore, for every x_1 , there are two equilibria corresponding to different values for x_2, x_3 .

Figure 1 depicts the utility gains at efficient equilibria,⁹ compared to the initial constrained allocation, and the flat lines represent the gain at the single-price equilibrium. Clearly, with

⁸Indeed, the equilibrium constraints require: $x_1 - x_1^* = -(x_2 - x_2^*)$, $p_1(x_1 - x_1^*) = -p_2(x_2 - x_2^*)$, and this double constraint entails $p_1 = p_2$.

⁹Inefficient equilibria are identified by checking the profile of utilities as a function of x_1 , and eliminating ranges in which the three countries' utilities are comonotonic with respect to x_1 .

multiple prices one cannot obtain a Pareto-improvement over the single-price equilibrium, but multiple prices allow for redistribution when permit endowments cannot be modified. The continuous curves and dashed curves correspond to the two roots of the equation for x_2 .

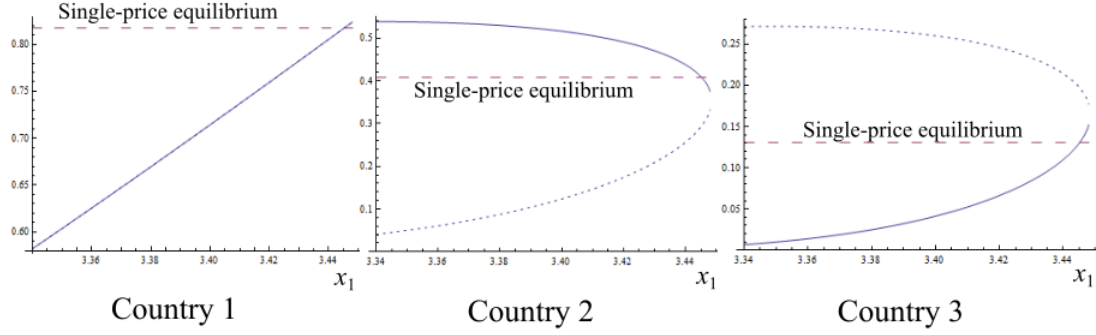


Figure 1: Utility gains from opening a differentiated-price permit market

The prices along these various equilibria move as shown in Fig. 2.

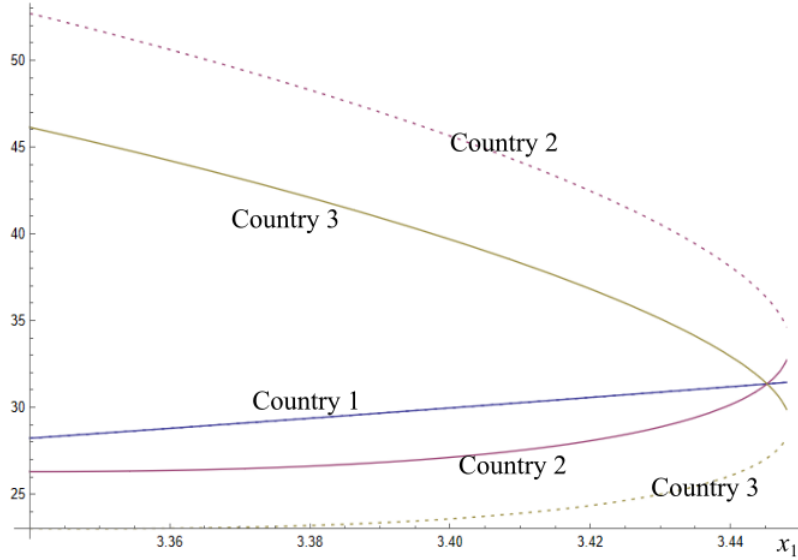


Figure 2: Individual prices in various market equilibria

As clearly shown in this illustration, a continuum of equilibria with differential prices is available, and the selection among this set would have to rely on the social welfare weights through the formula in Table 6. The concrete implementation of the chosen equilibrium could involve defining bonuses and penalties around a baseline price, for the different agents, and letting the market equilibrium be found by usual tâtonnement for this baseline price.

4 Fairness in burden-sharing: how to choose an allocation

Our main conclusion at this stage is that differential carbon prices can serve as a relevant distributive tool when constraints prevent direct transfers or appropriate distributions of permit allowances. Therefore, one cannot separate the question of the differentiation of carbon prices from the question of the fair allocation of burden-sharing. The two questions are inextricably linked, and should be addressed simultaneously. In the previous section, we left the question of which set of carbon prices and transfers should be chosen to the choice of welfare weights, where the weights appearing in the social welfare function were placeholders for unspecified distributive priorities. In this section we provide links between burden-sharing principles and implementation through appropriate social welfare weights.

However, an important difficulty emerges in this exercise. In some relevant contexts, certain equity principles cannot be directly implemented either because the available instruments are too limited in their range, or because the direct implementation would clash with efficiency requirements. We propose ways to get around such difficulties and obtain allocations which are reasonable in the considered context and are grounded in these burden-sharing principles. Some extensions take inspiration from the equivalence approach in fair allocation theory. This provides an array of possible applications of burden-sharing principles to the settings without transfers, with unlimited transfers and with market-constraint transfers we have studied in this paper.

4.1 No transfers

We know at this point that carbon prices should generally differ by country if transfers are absent or constrained. We have, however, left open how much carbon prices should differ, which is the policy-relevant question. To answer this question, we specify relevant inequalities for climate change mitigation and policy. Let $z_i, z_i^B, \bar{z}_i$ denote the actual, business-as-usual, and past historical emissions, respectively. Abatement is therefore equal to $x_i = z_i^B - z_i$. Total per capita¹⁰ emissions of a country, over recorded history and the regulated period, are equal to $z_i + \bar{z}_i$.

We now introduce different allocation rules embodying various equity principles. In addition to the following equity conditions, the BLS condition is always assumed to be fulfilled, and compatible with these additional conditions (i.e., there always exists an efficient allocation satisfying the conditions below). In some cases, there may be several allocations that satisfy the required conditions.

¹⁰We do not here address the question of deciding what the relevant population is, since different time scales could be chosen to apply our static model. In the numerical application presented in Section 5, we take the recent population size of the country as a proxy for the relevant size parameter.

The first rule is based on the principle that each country should contribute to the same extent to the public good on a per-capita basis.

Equal provision All abatement efforts $x_i = z_i^B - z_i$ are equal.

To find the allocation that satisfies the principle of Equal provision, one inserts the utilities $U_i(e_i - g_i(x_0), \sum n_j x_0)$ in the BLS condition and seeks the value of x_0 that fulfills the condition. The allocation $(x_0)_{i=1,\dots,n}$ then maximizes the social welfare function in Eq. 1 for a particular set of welfare weights. These welfare weights can be derived by inserting the allocation in the FOC from Eq. 2. We obtain (without normalizing the sum of weights):

$$\alpha_i = 1/U_{ic}g'_i(x_0),$$

where U_{ic} is evaluated at x_0 . Notice that the welfare weight that is implied by the Equal provision rule is higher for countries with lower marginal utility of consumption. Countries with a higher initial endowment thus receive a higher social welfare weight. The Equal provision rule may therefore be unattractive from a normative point of view, as it incorporates a “grandfathering” element.

The Equal provision rule may seem an intuitive approach for global public good provision. For climate change, however, the level of emissions is often more salient than abatement. We therefore introduce the following three rules that are based on actual, business-as-usual and historical emissions and that have been discussed in previous literature, public debates, or have been applied in international climate policy. Remember that variables are expressed in per capita terms in this paper.

Equal relative provision All abatement efforts $x_i = z_i^B - z_i$ are proportional to z_i^B .

Equal emissions All emission levels z_i are equal.

Historical responsibility All emission levels, including historical emissions, i.e., $z_i + \bar{z}_i$, are equal.

In the context of ethical decisions, climate change mitigation is often also related to the capacity of countries (“ability to pay”) to conduct large policy projects. We translate this approach in terms of our model by the following rule.

Progressive burden sharing All consumption levels are weakly increasing in endowment, but weakly less than proportional.

Progressive burden sharing encompasses a class of burden sharing rules. At one end of the spectrum, the consumption of countries would be exactly proportional to their endowment. Concretely, countries with a higher endowment (which we take as a measure for capacity) would have higher abatement costs (as costs have to be proportional to endowment). At the other end of the spectrum, Progressive burden sharing can require equal consumption between countries. That is, countries would abate until their consumption levels are equal. This case is quite extreme. In between both extremes, one may implement a rule in the spirit of the United Nations Right to Development, where some countries can develop without abatement obligations until they reach a certain development threshold, and abatement costs increase with endowment for the countries above that threshold.

For each of these rules, one again seeks to find the allocation that satisfies the principle and the BLS condition. Specifically, one inserts the following utilities in the BLS condition and seeks for the relevant variable:

- Equal relative provision: $U_i(e_i - g_i(rz_i^B), \sum n_j rz_j^B)$ seeking r ;
- Equal emissions: $U_i(e_i - g_i(z_i^B - z_o), \sum n_j (z_j^B - z_o))$ seeking z_o ;
- Historical responsibility: $U_i(e_i - g_i(\bar{z}_i + z_i^B - Z), \sum n_j (\bar{z}_j + z_j^B - Z))$ seeking Z ;
- Progressive burden sharing, consumption proportional to endowment: $U_i(re_i, \sum n_j g_j^{-1}((1-r)e_j))$ seeking r that defines each $x_i = g_i^{-1}((1-r)e_i)$;
- Progressive burden sharing, equal consumption: $U_i(c_0, \sum n_j g_j^{-1}(e_j - c_0))$ seeking c_0 that defines each $x_i = g_i^{-1}(e_i - c_0)$.

The welfare weights that enter the social welfare function which satisfies each principle are defined again through the first-order-conditions. Denote the allocation satisfying a rule from above as $x^* = (x_i^*)_{i=1,\dots,n}$. The corresponding welfare weights are

$$\alpha_i^* = 1/U_{ic} g_i'(x_i^*),$$

where U_{ic} is evaluated at x^* .

Example Let us illustrate the Progressive burden sharing rule, with proportional consumption, with the example introduced in Section 3. The BLS condition reads $\sum_i \frac{\theta_i c_i}{e^{\gamma_i x_i}} = 1$. Consumption is, for all i ,

$$c_i = e_i - g_i(x_i) = e_i(1 - r).$$

With $g_i(x_i) = \frac{e^{\gamma_i x_i} - 1}{\gamma_i}$, we thus have $e^{\gamma_i x_i} = 1 + \gamma_i e_i r$. The BLS condition becomes:

$$\sum_i \frac{\theta_i e_i (1 - r)}{\gamma_i e_i r + 1} = 1,$$

which then needs to be solved for the value of r to find the Progressive burden sharing allocation with proportional consumption.

In case there is no value of $r \in [0, 1]$ satisfying the above BLS equation, one can instead define proportions r_i for each i , through the equation

$$c_i = e_i (1 - r_i),$$

and maximize an inequality averse function such as $W = \frac{1}{1-\varepsilon} \sum_i (1 - r_i)^{1-\varepsilon}$ under the BLS constraint

$$\sum_i \frac{\theta_i e_i (1 - r_i)}{\gamma_i e_i r_i + 1} = 1.$$

The discussion so far has presented a theory of how to choose the weights by proposing different rules that are relevant in the climate change context. Which of the rules is appropriate to define the socially optimal allocation? To answer this question, the rules can be tested for different more basic ethical principles that they may fulfill or violate. We present such an approach in Appendix A, and refer the interested reader there.

4.2 Unlimited transfers

In the case in which transfers can be made freely between countries, there is paradoxically less freedom to adjust abatement efforts, but this is simply because efficiency then requires equalizing marginal costs, in addition to the BLS condition.

Nevertheless, the fair allocation rules introduced in the previous subsection can still be applied indirectly. One can indeed seek allocations which are Pareto-equivalent to allocations in which the relevant equality or proportionality conditions are satisfied. This approach takes inspiration from the equivalence approach that is a cornerstone of the theory of fair allocation (e.g., Moulin 1987).

For instance, consider Equal emissions. It is generally impossible to equalize $z_i = z_i^B - x_i$ across countries when $g'_i(x_i)$ is also equalized. But one can seek an efficient allocation such that, for some z_0 , one has, for all i ,

$$U_i(c_i, X) = U_i(e_i - g_i(z_i^B - z_0), X^*)$$

where X^* is some reference level of the public good. By adjusting the level of z_0 up or down, one moves all the utilities of the countries in sync, and one simply has to seek the point where this distribution of utility levels is efficient. Observe that the allocation of abatement $(z_1^B - z_0, \dots, z_N^B - z_0)$ is not in general compatible with the level X^* , i.e., in general one has $X^* \neq \sum_i z_i^B - z_0$. The contemplated allocation is hypothetical but nevertheless relevant because it yields a distribution of utilities that embodies the tolerance for inequalities in endowments and abatement cost functions that motivates the Equal emission rule.

The choice of X^* is of some importance, because it does affect the selection of the allocation. We propose to pick a high level, corresponding to the typical climate of the Holocene, for the following reason. A high level of the public good guarantees that, among two countries with the same characteristics e_i, g_i , when $X < X^*$ the country that cares more about the climate in its utility function U_i would be granted a greater consumption level in the actual allocation, in recognition of the greater disadvantage the failure to achieve X^* represents for it.¹¹

Feasibility issues may occur for this approach, because the equivalent allocations of the desired form, such as $U_i(e_i - g_i(z_i^B - z_0), X^*)$, may not cover a large enough range of welfare distributions. For instance, one requires $x_i = z_i^B - z_0 \geq 0$ for all i , but also $g_i(z_i^B - e_0) \leq e_i$ in order to keep $c_i \geq 0$. Assuming that the range of z_0 is then

$$A_0 = \left[\min_i z_i^B - g_i^{-1}(e_i), \min_i z_i^B \right],$$

the set

$$\left\{ (U_i(e_i - g_i(z_i^B - z_0), X^*))_{i=1, \dots, N} \mid z_0 \in A_0 \right\}$$

may be too high and fail to intersect with the Pareto frontier. There are ways to deal with this problem.

First, one may allow equivalent emissions to go above the unregulated level to eliminate the upper bound in A_0 . A country i with $z_0 > z_i^B$ in the equivalent utility would not have any costs of abatement ($g_i \equiv 0$) but would receive a subsidy for emissions to let equivalent utility increase in emissions: $U_i(c_i, X^*) = U_i(e_i + \sigma_i \cdot (z_0 - z_i^B))$.

Second, one may allow for a degree of inequality instead of seeking a common z_0 . This is done by defining a personal value for the equivalent level of emissions through the equation

$$U_i(c_i, X) = U_i(e_i - g_i(z_i^B - v_i), X^*).$$

The range of utilities covered by this approach is much greater, since for every i , the range of

¹¹An extensive discussion of the choice of a high reference for public goods, in this approach, can be found in Fleurbaey and Blanchet (2013).

v_i is

$$A_i = [z_i^B - g_i^{-1}(e_i), z_i^B],$$

so that utility goes from $U_i(0, X^*)$ up to $U_i(e_i, X^*)$. The Cartesian product of such utility ranges over the whole population is most likely to intersect with the Pareto frontier. One can then maximize a function such as $W = \frac{1}{1-\varepsilon} \sum_i (a + v_i)^{1-\varepsilon}$ under the constraints of the BLS condition and equality of marginal abatement costs.

Example Let us apply the Equal provision rule to the example introduced in Section 3, with unlimited transfers. The important equation reads, for all i ,

$$\ln c_i + \theta_i X = \ln(e_i - g_i(x_0)) + \theta_i X^*,$$

where $c_i = e_i + t_i - g_i(x_i)$.

Together with the conditions for optimality $\sum_i t_i = 0$ and $g'_i(x_i) = p$ (for some uniform p), these equations define transfers and contributions as functions of x_0 . More precisely, one has

$$t_i = -e_i + \frac{p-1}{\gamma_i} + (e_i - g_i(x_0)) e^{\theta_i \left(X^* - \ln p \sum_j \frac{n_j}{\gamma_j} \right)}$$

so that the constraint

$$\sum_i t_i = 0 = \sum_i \left[-e_i + \frac{p-1}{\gamma_i} + (e_i - g_i(x_0)) e^{\theta_i \left(X^* - \ln p \sum_j \frac{n_j}{\gamma_j} \right)} \right]$$

determines p as a function of x_0 , and then $x_i = \frac{\ln p}{\gamma_i}$ as well. Denote the contributions as $x_i(x_0)$ and in aggregate $X(x_0)$.

The BLS condition is $\sum_i \frac{\theta_i c_i}{p} = 1$ and inserting it yields, for every j ,

$$\sum_i \theta_i [(e_i - g_i(x_0)) e^{\theta_i (X^* - X(x_0))}] = g'_j(x_j(x_0))$$

which needs to be solved for the value of x_0 to find the socially optimal allocation with equal provision under free international transfers.

The other equity principles can be implemented in the same way, taking the same formulas as in the previous subsection, but applied to the hypothetical (Pareto-equivalent) allocation rather than the actual allocation.

4.3 Market-constrained transfers

When a carbon market implements transfers in a way that is partly determined by initial endowments with permits and partly operated through differentiated prices, Equation (5) operates as a similar constraint as the equality of marginal costs in the previous subsection. But the Pareto-frontier delineated by the market-constrained form of redistribution is generally quite limited in scope, which makes it even more difficult to relate possible selections from this set to burden-sharing principles.

One can again extend the equivalence method along the same lines as in 4.2, and hope that the range of attainable utilities intersects with the Pareto frontier. Consider Equal consumption. One then measures the personal equivalent value v_i by solving the equation

$$U_i(c_i, X) = U_i(v_i, X^*)$$

and one can then plug the distribution of v_i into a social welfare function that is inequality averse and increasing in the v_i s (e.g., $W = \frac{1}{1-\varepsilon} \sum_i v_i^{1-\varepsilon}$ for $\varepsilon > 0$). The range of attainable utilities goes from $U_i(0, X^*)$ to $U_i(+\infty, X^*)$ and is thus relatively large.

It remains to bridge the gap between these new social welfare functions and the weighted sum of utilities that underlies the computation of the MSV_i featured in equation (5). We will do it for two examples and a general social welfare function $\sum_i \psi(v_i)$.

Equal emissions: Equivalent utility reads $U(c_i, X) = U(e_i - g_i(z_i^B - v_i), X^*)$. The MSV_i under $\sum_i \psi(v_i)$ is equal to

$$MSV_i = \psi'(v_i) \frac{n_i U_{ic}}{U_{ic}^* g'_i(z_i^B - v_i)}$$

(where U_{ic}^* is the marginal utility of consumption in the hypothetical allocation). Thus, equating this with the $MSV_i = \alpha_i U_{ic}$ yields

$$\alpha_i = \frac{\psi'(v_i)}{n_i U_{ic}^* g'_i(z_i^B - v_i)}.$$

Equal consumption: With the same method one computes

$$MSV_i = \psi'(v_i) \frac{U_{ic}}{n_i U_{ic}^*},$$

$$\alpha_i = \frac{\psi'(v_i)}{n_i U_{ic}^*}.$$

The difference between the MSV_i for the two rules captures the fact that equal consump-

tion gives priority to those with low consumption in the hypothetical allocation, whereas equal emissions gives priority to those with low consumption and low marginal costs in the hypothetical allocation.

Example Let us apply the Equal consumption rule to the example introduced in Section 3, in the market-constrained case. For the social welfare function $\sum_i \psi(\cdot)$, we adopt the isoelastic formula again, and obtain the program:

$$\begin{aligned}
& \max_{v,p,x} \sum_i \frac{v_i^{1-\eta}}{1-\eta} \\
& \text{s.t.} \quad \ln(c_i) + \theta_i X = \ln(v_i) + \theta_i X^*, \quad \forall i \\
& \quad c_i = e_i - \frac{e^{\gamma_i x_i} - 1}{\gamma_i} + p_i (x_i - x_i^*), \quad \forall i \\
& \quad X = \sum_i x_i, \\
& \quad x_i = \frac{1}{\gamma_i} \ln p_i, \quad \forall i \\
& \quad \sum_i p_i (x_i - x_i^*) = 0, \\
& \quad \sum_i x_i = \sum_i x_i^*.
\end{aligned}$$

5 Estimates of burden sharing

This section uses a calibrated numerical model to quantify our theoretical results. We consider all transfer cases – no transfers, unlimited transfers and transfers within a carbon market – and implement the theoretical model for four equity principles introduced above: Equal consumption, Costs proportional to GDP, Equal emissions and Equal relative provision. We find that generally low-income countries tend to have a lower burden compared to high-income countries to achieve the equity principles, either through lower carbon prices or higher transfers received, or both. For all scenarios, we show that region-specific carbon prices and/or transfers improve social welfare compared to a globally uniform price.

5.1 Description of numerical model and implementation

Our numerical model is based on RICE50plus v2.0.0 (Gazzotti et al., 2021). RICE50plus divides the world into 57 countries and regions (see list of regions in Appendix B), with each regions economy similar to the DICE model (Nordhaus, 2017) but based on SSP scenario projections (Riahi et al., 2017). Climate change interacts with the economy via climate damages, based on regional temperature and recent damage estimates (Van Der Wijst et al., 2023), and via costs of reducing emissions. For a description of the model see Gazzotti et al. (2021)

and the open-access code available at <https://github.com/witch-team/RICE50xmodel>. Appendix C describes the technical details of our model and how each scenario is implemented.

We derive a static model from the dynamic RICE50plus in the following way. First, we calibrate endowment (equal to GDP), abatement costs and unregulated emissions for the year 2025 from the corresponding values of RICE50plus. Second, benefits from reducing emissions in 2025 are the regional social cost of carbon derived from shadow prices of emissions and consumption in each region (see Appendix C for details). The regional social cost of carbon, for each region, measures the marginal value of avoiding one more unit of emissions in 2025 and thus consuming more in the future, versus consuming one more unit in 2025 and thus not abating.

Regional utility has the following form:

$$U_i = n_i \log(c_i + b_i), \forall i$$

Regional consumption is $c_i = e_i - g_i(x_i) + t_i \cdot m2p_i$, where e_i is per capita GDP of regions in 2025. The terms $g_i(x_i)$ are regional per capita abatement costs as a function of regional per-capita abatement $x_i \geq 0$.¹² Benefits from abatement b_i are total abatement X multiplied with regional social cost of carbon estimates, normalized by population. The globally aggregated social cost of carbon is 80 USD (PPP)/tCO₂. Regional e_i , g_i and b_i are measured in Purchasing Power Parity (PPP) 2005 USD. For the scenarios with transfers, per capita transfers t_i are converted from Market Exchange Rates (MER) to PPP using the conversion rates $m2p_i$. Transfers add up to zero in all scenarios: $\sum_i n_i t_i = 0$, where n_i denotes each region's population. A positive transfer $t_i > 0$ means region i receives a payment, a negative transfer $t_i < 0$ means region i makes a payment. Transfers are equal to permit trade at region-specific carbon prices when a carbon market is implemented.¹³

Carbon prices and transfers are estimated for all transfer cases: no transfers, unlimited transfers and carbon market transfers. In case of unlimited transfers, the efficient carbon price for all regions is 50 USD (MER)/tCO₂, which is the globally aggregated social cost of carbon in MER units. For carbon market transfers, we choose the total amount of permits equal to total emissions under an equal carbon price of 50 USD (MER)/tCO₂ for all regions. We test two initial distributions of permit endowments: grandfathered permits and an equal-per-capita endowment. For each transfer case, four equity principles are implemented as described in Section 4: Equal consumption, Costs proportional to GDP, Equal emissions, and Equal relative provision.

To measure how much differentiated carbon prices and/or transfers improve equity in burden

¹²We do not assume an upper bound on abatement, allowing for negative emissions in all regions.

¹³Transfers in the carbon market are: $t_i = g'_i(x_i)/m2p_i \cdot (x_i - x_i^*)$, where x_i^* is the initial endowment with permits (see Appendix C for details).

sharing, we compare social welfare at the allocation that fulfills an equity principle with social welfare at the allocation with equal carbon prices. To that end we convert social welfare in utility units to USD values using the equally distributed equivalent. The increase in social welfare ΔSWF derived below approximately measures how much money each person would have to receive at the equal carbon price solution for society to be equally well off as when achieving an equity principle with differentiated carbon prices and/or transfers. We proceed in four steps to compute ΔSWF .

First we select the efficient allocation with a globally uniform carbon price as the reference to compute gains in social welfare. The reference case differs between the transfers cases:

1. No and unlimited transfers: the reference case is the allocation at the Pareto-efficient frontier with a globally uniform price at 50 USD (MER)/tCO₂ and no transfers.
2. Transfers in a carbon market: the reference case is the efficient allocation with a single carbon price at 50 USD (MER)/tCO₂ and trade in permits that clears the permit market (the usual carbon market equilibrium, e.g. in Flachsland et al. 2009). The reference allocation is different for the two initial endowments with permits.

Denote this allocation $c_i^{ep} + b_i^{ep}$, bearing in mind that it is different across the transfer cases.

Second we consider each equity principle under each transfer case, and denote $c_i^* + b_i^*$ as the (constrained) efficient allocation that fulfills the principle. Social welfare is: $SWF^* = \sum_i \alpha_i n_i \log(c_i^* + b_i^*)$, where the welfare weights α_i are endogenous to the specific equity principle (Equal consumption, ...) and assumption on transfers (No transfers, ...), as derived in the previous section. Figure 6 in the Appendix reports the welfare weights for our numerical model.

Third, we compute the level of social welfare for each equity principle at the solution with a globally uniform price: $SWF^{ep} = \sum_i \alpha_i n_i \log(c_i^{ep} + b_i^{ep})$. The welfare weights α_i are those computed as explained in the previous paragraph, and are again specific to each equity principle and assumption on transfers.

Fourth, we convert both levels of social welfare to the equally distributed equivalent welfare. The equally distributed equivalent welfare is the level of equal consumption across regions that gives rise to the same level of social welfare. For our numerical model it is:

$$\begin{aligned} \sum_i \alpha_i n_i \log(c^{ede}) &= SWF \\ \Rightarrow c^{ede,*} &= \exp\left(\frac{SWF^*}{\sum_i \alpha_i n_i}\right) c^{ede,ep} = \exp\left(\frac{SWF^{ep}}{\sum_i \alpha_i n_i}\right). \end{aligned}$$

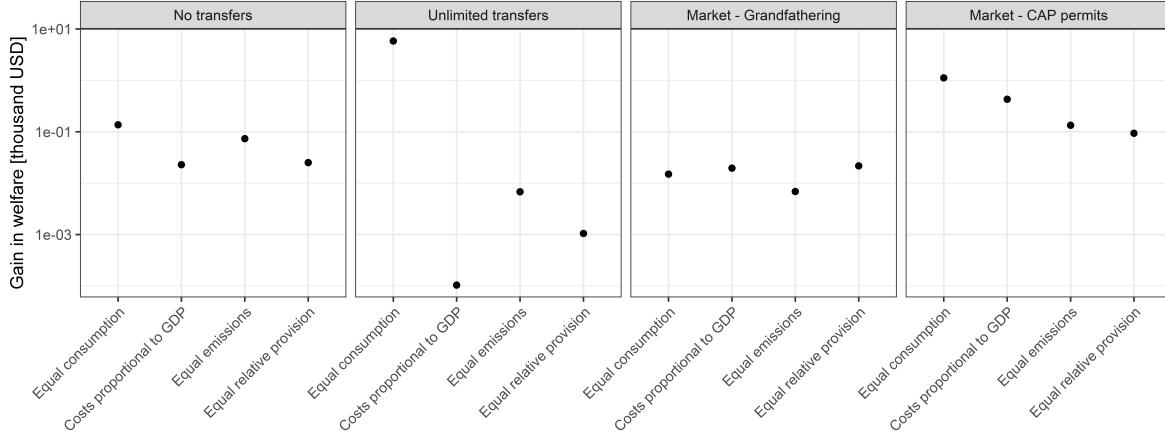


Figure 3: Increase in social welfare compared to the case of equal carbon prices, measured by the equally distributed equivalent (see Eq. 6). Panels show the transfer cases: no transfers, unlimited transfers, and carbon market transfers with initial permits either grandfathered or based on equal-per-capita-emissions. The transfer cases are solved for four equity principles: Equal consumption, Costs proportional to GDP, Equal emissions and Equal relative provision. The gain is relative to the Pareto-efficient allocation with equal prices and no transfers for the cases of no transfers and unlimited transfers and relative to the allocation with equal prices and permit market clearing for the case of carbon market transfers.

The difference in social welfare:

$$\Delta SWF = c^{ede,*} - c^{ede,ep}. \quad (6)$$

measures the gain from differentiated carbon prices and/or transfers compared to a uniform price.

5.2 Results

Figure 3 shows the gain in social welfare with region-specific carbon prices (no transfer case), a common carbon price and transfers (unlimited transfers) and a combination of region-specific prices and corresponding transfers in a carbon market (carbon market transfers). All welfare gains are positive, i.e. the reference case of a globally uniform carbon price corresponds to none of the four equity principles we consider. We next discuss the results for each transfer case and equity rule in detail.

No and unlimited transfers Considering the cases of no transfers and unlimited transfers, there is a clear ranking by how much differentiated carbon prices or transfers increase social welfare: the gain is largest for Equal consumption, followed by Equal emissions, Equal relative provision and Costs proportional to GDP. The intuition is clear: consumption and emissions at

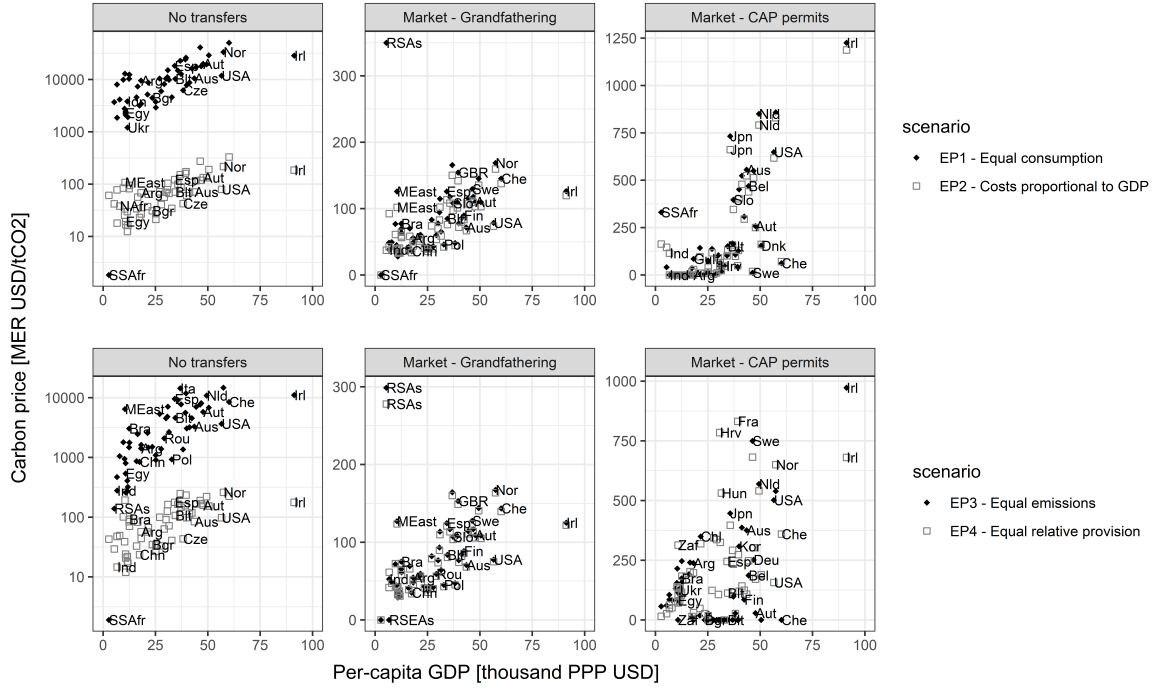


Figure 4: Carbon prices in Market Exchange Rates against GDP in Purchasing Power Parity for different scenarios. Panels show the no-transfer case and two cases of carbon market transfers, with initial permits either grandfathered or based on equal-per-capita-emissions. The three cases are solved for four equity principles: Equal consumption, Costs proportional to GDP, Equal emissions and Equal relative provision.

the equal price reference allocation without transfers are quite different across regions. Thus, using differentiated carbon prices or transfers to equalize consumption or emissions allows sizable gains in social welfare. The reference allocation with equal prices equates marginal abatement costs and is therefore closer to an allocation that equalizes relative provision and costs to GDP ratios across regions. Gains in social welfare from using differentiated carbon prices or transfers are thus lower.

As consequence, carbon prices to achieve Equal relative provision and Costs proportional to GDP have a moderate range in case of no transfers (roughly between 1 and 300 USD (MER)/tCO₂, Figure 4). Equivalently, transfers are moderate to achieve both equity principles in case of unlimited transfers (below 2 percent of GDP received or paid, Figure 5).

The equity principles Equal Consumption and Equal emissions are different: the range of carbon prices (from 1 to 50000 USD/tCO₂) and transfers (from -80 to +600 percent of GDP) needed to equate consumption or emissions across countries is extreme. (Note that extreme carbon prices and transfers are just illustrative as the results go beyond what the RICE50plus model is calibrated for.) Extreme prices and transfers are necessary because

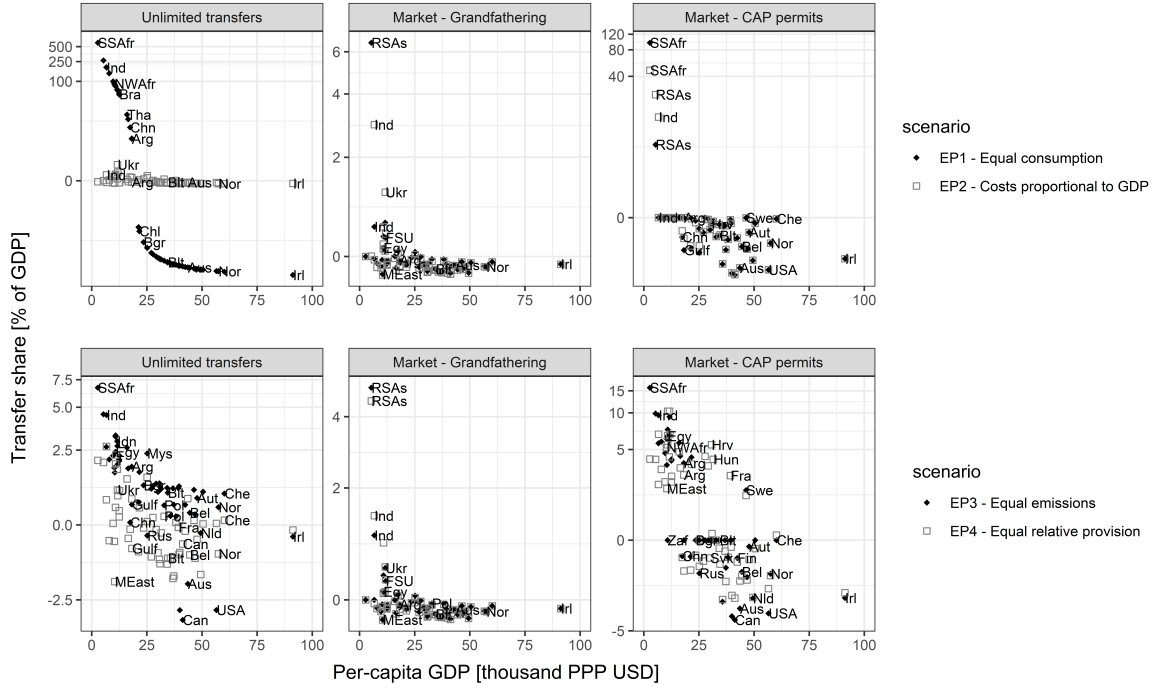


Figure 5: Transfers as share of GDP against GDP in Purchasing Power Parity for different scenarios. Panels show the unlimited transfer case and two cases of carbon market transfers, with initial permits either grandfathered or based on equal-per-capita-emissions. The three cases are solved for four equity principles: Equal consumption, Costs proportional to GDP, Equal emissions and Equal relative provision.

pre-existing differences in consumption and emissions are extreme, spanning orders of magnitude. Country-specific prices at this extreme range mean large total abatement costs so that welfare gains are limited (at 140 and 70 USD for Equal consumption and Equal emission, respectively). For example, under no transfers, every person only consumes 2800 USD when country-specific carbon prices equalize consumption to the value of the lowest-income region. Note that this allocation is constrained Pareto-efficient. In case of unlimited transfers, every person consumes 19200 USD and welfare increases by more than 5000 USD per person because a uniform carbon price of 50 USD (MER)/tCO₂ allows for the largest amount of aggregate consumption that can be redistributed. Transfers address pre-existing inequalities in endowments more effectively than differentiated carbon prices.¹⁴

¹⁴Considering the Equal emission principle, gains in social welfare are higher in case of no transfers (70 USD) than with unlimited transfers (7 USD). This is the case because emissions are only actually equalized in case of no transfers, while with unlimited transfers emissions are equalized in equivalent terms (see Section 4.2 and Appendix C). Equalizing equivalent emissions requires mostly moderate transfers as transfers result from different (equivalent) abatement costs across regions, which are usually a small fraction of GDP (transfers are in the range of -4 to +7 percent of GDP, Figure 5). The reference case with a uniform price and no transfers is thus closer to the socially optimal allocation in case of unlimited transfers than in case of no transfers.

Carbon market transfers When transfers are constrained through permit trading, the initial endowment with permits determines by how much the reference case with a uniform carbon price and permit market clearing deviates from an equity principle. The initial endowment with permits also determines to a certain extent the magnitude and direction of transfers. We test two initial permit allocations. For both, the total amount of permits is equal to emissions under a common carbon price of 50 USD (MER)/tCO₂. In the first grandfathering approach, regions receive an initial endowment with permits proportional to their unregulated emissions. In the second equal per-capita permit allocation, regions receive initial permit endowments proportional to their population.

The four equity principles were implemented using the equivalence approach and maximizing an inequality-averse social welfare function in the respective variable (consumption, costs to GDP ratio, emissions, relative provision; see Section 4.3 and Appendix C). The maximization is a local optimum that improves upon the reference allocation with equal prices and permit market clearing.

With grandfathered permits, carbon prices that differ only moderately across regions (0 to 400 USD (MER)/tCO₂, Figure 4) and associated rather moderate transfers from permit trade (between 0.5% of income paid and 7% of income received, Figure 5) lead to moderate gains in social welfare (7 to 22 USD, Figure 3). Grandfathering permits prevent large increases in social welfare for two reasons. First, the reference case with a uniform price and permit market clearing is close to an allocation that equalizes costs to income ratios and relative provision, hence region-specific prices cannot improve social welfare by a lot. Second, grandfathering permits can prevent redistribution in a way that improves social welfare. To see why, take the Equal consumption equity principle and the example of low-income region Sub-Saharan Africa (SSAfr). Sub-Saharan Africa receives a small initial permit endowment due to its low unregulated emissions. With this initial allocation Sub-Saharan Africa would be required to buy permits and thus pay a transfer at the equal price reference allocation. At the allocation under the Equal consumption equity principle, the carbon price and transfers of Sub-Saharan Africa are zero, which contributes to the equity principle compared to transfers paid. However, society seeks to increase consumption in Sub-Saharan Africa which grandfathered permits hinder.

Social welfare increases considerably with equal-per-capita endowments with permits: over 1000 USD in case of Equal consumption and above 90 USD for the other three equity principles (Figure 3). The reason is two-fold. First, equal-per-capita endowments with permits allow significant transfers, and thus enable larger changes in the final allocation. Second, the uniform price reference scenario with permit market clearing significantly deviates from all four equity principles.¹⁵ Take the example of the Equal consumption principle: low-income countries

¹⁵This is also the case for the Equal emissions principles. Handing out permits on an equal-per-capita basis

receive a high initial endowment with permits compared to unregulated emissions. Thus higher carbon prices in low-income countries increase their transfer received from permit exports, improving burden sharing significantly. Across all equity principles, carbon prices show a considerable variation across regions (0 to 1200 USD (MER)/tCO₂) and allow substantial transfers between countries (up to 96% of income received).

6 Conclusions

Country-specific carbon prices can play a significant role in international climate governance. If constraints limit the scope of transfers between countries, country-specific carbon prices are efficient, i.e. (almost all) allocations at the constrained Pareto-frontier without transfers feature unequal carbon prices. Allocations in carbon markets are no exception to this rule, if countries have fixed initial permit endowments: country-specific permit prices are efficient on a constrained Pareto-frontier. The international community may thus rely on country-specific carbon prices – also within carbon markets – to address equity concerns in burden sharing while combating climate change at low costs. The popular idea of a common global carbon price is efficient and optimal only when transfers are freely available to deal with pre-existing inequalities or the prevailing level of inequality is optimal.

We characterize how country-specific prices in a carbon market should be designed to promote equity. As a rule of thumb, we find that a country has a higher optimal carbon price than the average in the market if two conditions are fulfilled. First, the country should have social priority based on the equity principle considered. Second, the country is able to abate above its fixed initial endowment with permits so that it can sell permits on the market. Alternatively, a high price also applies if the country buys emission permits and has a lower-than-average social priority.

Our results indicate that equity may demand extreme carbon prices for some countries or large transfer payments between countries to offset unfair burdens under a common global carbon price. Combining moderate country-specific carbon prices with moderate transfers – in an international price-and-transfer regime or in an international carbon market – may thus be best to address equity and efficiency while remaining politically feasible.

We wish to highlight several limitations of our model that should be relaxed in future research. First, the static setting in this paper should be extended to a dynamic framework, which is key

leads to transfers that often do not correspond to equal emissions in equivalent utility. Take the example of Canada that has high regulated emissions at a price at 50 USD(MER)/tCO₂ (13 tCO₂ per capita). Under trade at a uniform price, Canada would have to pay 1.2% of its GDP in permit imports as it receives a low number of permits. However, the transfers paid do not offset Canada's high regulated emissions. Increasing the Canadian price to 390 USD(MER)/tCO₂ leads to Canada paying above 4% of their GDP in permit imports, improving equity by reducing inequality in equivalent emissions.

to burden sharing in climate change mitigation. Second, we abstract from inequality below the national level, which could be an important component to carbon pricing if redistribution is limited. Third, we do not consider risky environments and how carbon prices should be designed to deal with uncertainty.

Appendix A: Ethical underpinnings of allocation rules in Section 4.1

The literature on burden-sharing generally offers allocation rules directly, without looking at the logical and ethical underpinnings of such rules beyond the motivations that are intuitively apparent in the formulation of the rules themselves. An important exception is the axiomatic analysis by de Villemeur and Leroux (2011), who, however, focus on how to share the monetary cost of climate damages rather than the allocation of emission rights. We do not provide axiomatic characterizations here but check which rules satisfy a number of key properties that relate to equity issues, in the context of the constrained model with no transfers and no market. Our analysis is summarized in Table 7, and the axioms are defined afterward.

	Eq.Provis.	Eq.Rel.Prov.	Eq.Emiss.	Hist.Resp.	Prog.Burd-Shar.
Impartiality	✓	✓	✓	✓	✓
Ind.SReliance	✓	✓	✓	✓	X
Em.Irresp.	✓	X	X	X	✓
Fut.Em.Resp.	X	X	✓	✓	X
Tot.Em.Resp.	X	X	X	✓	X

Table 7: Properties of the different allocation rules

Impartiality is a most basic condition, stipulating that two countries with the same characteristics should have the same abatement effort. This is satisfied by all rules considered in this paper.

Impartiality When countries have identical characteristics, then for any selected allocation there is x_0 such that $x_i = x_0$ for all i .

The next axiom imposes a strong indifference to inequalities: if countries differ in their endowments, abatement costs or sensitiveness to the public good, this will not influence which allocation will be chosen by the social welfare function. Perhaps surprisingly, most of the burden-sharing rules considered in the literature do satisfy this axiom.

Individual self-reliance If the allocation of abatement x is selected, and then the profile of U_i, e_i, g_i changes for some countries i , but x remains Pareto efficient, then x remains selected.

Testing for Individual self-reliance in Table 7 reveals that all rules that only base the allocation on characteristics related to emissions fail to acknowledge any background inequalities and even vulnerability to climate change. Individual self-reliance is dealing with individual characteristics which do not generate externalities on other countries when they change.

When considering background inequalities such as endowments and utilities as issues outside of decisions on climate change, picking the Equal provision, Equal relative provision, Equal emission, or Historical responsibility rule seems attractive. However, these rules also neglect differences in abatement costs and vulnerability to climate change for picking an allocation, which seems unattractive as countries may not be held responsible for, at least, their vulnerability to climate change. In this case, the Progressive burden sharing rule seems more appropriate.

The next axiom embodies total leniency to business-as-usual behavior as well as historical responsibilities.

Emission irresponsibility If the allocation of abatement x is selected, and then the profile of $z_i^B, \bar{z}_i$ changes for some countries i , but x remains Pareto efficient, then x remains selected.

In contrast to Individual self-reliance, Emission irresponsibility is about parameters which directly affect other countries through climate change. Not changing the abatement requested from a country when its emissions characteristics change is tantamount to relieving it of its climate change responsibilities toward the rest of the world. In this respect, Equal provision and Progressive burden sharing appear quite unattractive for climate change mitigation: they fail to acknowledge that countries cause harm to the whole world when their unregulated emissions increase. It may therefore not be surprising that the rules that depart from Emission irresponsibility have been discussed often in the literature and public debate on climate change: Equal relative provision and Equal emissions (known in the context of the Kyoto Protocol and the Paris Agreement) as well as Historical responsibility. But it is striking that the other two rules introduced above do satisfy this principle.

It is noteworthy that the rules that violate Emission irresponsibility are also the rules that fulfill Individual self-reliance. If one wants to relieve countries from contributing much to the public good if their endowments are low, the Progressive burden sharing rule is better suited, with the trade-off that it fulfills Emission irresponsibility.

To distinguish between the rules that eschew Emissions irresponsibility, the next two principles are introduced, which hold countries responsible for their future or the sum of their future and past emissions.

Future emission responsibility If the allocation of emission z is selected, and then the profile of z_i^B changes for some countries i , but z remains Pareto efficient, then z remains selected.

Total emission responsibility If the allocation of emission z is selected, such that total emissions are equal to $\bar{z} = z + \bar{z}$, and then the profile of $z_i^B, \bar{z}_i$ changes for some countries

i , but keeping the same $\bar{z}$ remains Pareto efficient, then the new allocation has emissions equal to the new value of $\bar{z} - \bar{z}$.

Unsurprisingly, these two properties are only fulfilled by the rules that base the allocation on the emission characteristics of countries. If one believes that countries should be responsible for the climate change that their past emissions caused, Historical responsibility could be chosen. If one only deems future emissions important, the Equal emissions approach is also justified. Of course, past emissions may be restricted to a time-horizon in which anthropogenic climate change was public knowledge.

In conclusion, if one is interested in finding a rule that is sensitive to inequalities due to endowments and vulnerability to climate change, as well as the current and past emissions record, one has to look outside of the class of equity rules considered in this paper, and probably outside the range of rules proposed in the literature. We leave this intriguing question for future research.

Appendix B: List of regions in numerical model

Region	Description	ISO countries
Arg	Argentina	ARG
Aus	Australia	AUS
Aut	Austria	AUT
Bel	Belgium	BEL
Bgr	Bulgaria	BGR
Blt	Baltic states	EST, LTU, LVA
Bra	Brazil	BRA
Can	Canada	CAN
Che	Switzerland	CHE
Chl	Chile	CHL
Chn	China	CHN
Cze	Czech Republic	CZE
Deu	Germany	DEU
Dnk	Denmark	DNK
Egy	Egypt	EGY
Esp	Spain	ESP
Fin	Finland	FIN
Fra	France	FRA
FSU	Former Soviet Union	ARM, AZE, BLR, GEO, KAZ, KGZ, MDA, TJK, TKM, UZB
GBR	UK	GBR
Gulf	Gulf Countries	ARE, BHR, IRN, IRQ, KWT, OMN, QAT, SAU, YEM
Grc	Greece	GRC
Hrv	Croatia	HRV
Hun	Hungary	HUN
Idn	Indonesia	IDN
Ind	India	IND
Irl	Ireland	IRL
Ita	Italy	ITA
Jpn	Japan	JPN
Kor	Korea	KOR
MEast	Middle East	ISR, JOR, SYR, LBN, PSE
Mex	Mexico	MEX

Mys	Malaysia	MYS
Nld	Netherlands	NLD
NAfr	North Africa	ESH, TUN, MAR
NWAfr	North-West Africa	LBY, DZA
Nor	Norway	NOR
Ocean	Pacific Island	CXR, COK, HMD, NFK, NIU, NRU, PCN, TKL, TUV, UMI, WLF, FJI, PNG, FSM, GUM, ASM, TLS, PYF, KIR, MNP, MHL, NCL, PLW, WSM, SLB, TON, VUT, NZL, TTO,
Pol	Poland	POL
Prt	Portugal	PRT
RCAm	Rest Central America	BES, CUW, SXM, ABW, BHS, BLZ, BRB, CRI, CUB, DMA, DOM, GRD, GTM, HND, HTI, JAM, LCA, NIC, PAN, SLV, VCT, BMU, SGS, TCA, VGB, VIR, AIA, ATG, BLM, CYM, GLP, KNA, MAF, MSR, MTQ, PRI
REur	Rest Europe	CYP, LUX, MLT, LIE, GRL, ISL, FRO, ALA, AND, GGY, GIB, IMN,JEY,MCO, SJM, SMR, VAT, SPM, BIH, ALB, MKD, MNE,SRB,KSV
Rou	Romania	ROU
RSAm	Rest South America	BOL, COL, ECU, FLK, GUF, GUY, PER, PRY, SUR, URY, VEN
RSAs	Rest South Asia	AFG, BGD, BTN, LKA, MDV, NPL, PAK
RSEAs	Rest South-East Asia	BRN, CCK, KHM, LAO, MMR, PHL, SGP, PRK, HKG, MAC, TWN, MNG
Rus	Russia	RUS
SSAfr	Sub-Saharan Africa	AGO, BEN, BWA, BFA, BDI, CMR, CPV, CAF, TCD, COM, COG, COD, CIV, GNQ, ERI, ETH, GAB, GMB, GHA, GIN, GNB, KEN, LSO, LBR, MDG, MWI, MLI, MRT, MUS, MYT, MOZ, NAM, NER, NGA, REU, RWA, STP, SEN, SYC, SHN, SLE, SOM, SSD, SDN, SWZ, TZA, TGO, UGA, ZMB, ZWE, DJI, IOT, BVT, ATF, COG, COD, CIV, GNQ, ERI, ETH, GAB, GMB, GHA, GIN
Slo	Slovenia	SVN

Svk	Slovakia	SVK
Swe	Sweden	SWE
Tha	Thailand	THA
Tur	Turkey	TUR
Ukr	Ukraine	UKR
USA	USA	USA
Vnm	Vietnam	VNM
Zaf	South Africa	ZAF

Appendix C: Detailed description of the numerical model

We set the RICE50plus model to the SSP2 scenario, which projects regional economic activity, population, and unregulated emissions. Emissions lead to temperature increases that are region-specific. Increased temperatures lead to reduction in output, where the damage function is estimated in Van Der Wijst et al. (2023). RICE50plus is solved as a non-cooperative Nash-equilibrium in emissions, where regions individually abate their emissions as an optimal response to climate damages that affect only their own economy, holding emissions of all other regions fixed.

In our static model, regional utility in 2025 is:

$$U_i = n_i \log \left(\frac{C_i + B_i}{n_i} \right)$$

Regionl population n_i is measured in million. Regional consumption is:

$$C_i = Y_i - G_i(x_i) + T_i \cdot \text{m2p}_i,$$

where Y_i is GDP of regions in 2025. $G_i(x_i)$ are regional abatement costs as a function of regional per-capita abatement $x_i \geq 0$. Regional abatement costs follow the functional form

$$G_i(x_i) = mx_i \left[\frac{1}{2} a_i (x_i/z_i^B)^2 + \frac{1}{5} b_i (x_i/z_i^B)^5 \right] \frac{z_i^B n_i}{1000} \quad (7)$$

with mx_i , a_i and b_i parameters and z_i^B unregulated per-capita emissions in the SSP2 scenario. Abatement is measured in GtCO₂. Y_i and G_i are measured in Trillion PPP 2005 USD. For the scenarios with transfers, transfers T_i are converted from Market Exchange Rates to PPP using the conversion rates m2p_i. Transfers add up to zero in all scenarios: $\sum_i T_i = 0$.

Regions derive the following benefits from abatement:

$$B_i = scc_i \sum_j \frac{x_j n_j}{1000}, \quad (8)$$

with scc_i regional social cost of carbon in 2005 PPP USD/tCO₂. The conversion factor 1000 ensures that variable units match. We derive regional social cost of carbon values from RICE50plus by using shadow prices along the non-cooperative Nash-equilibrium in emissions. For each region, the shadow prices are derived from maximizing each region's total discounted utility, holding all other region's abatement levels fixed. Regions internalize their own damages from additional emissions by optimally adjusting abatement. Total discounted utility is derived from isoelastic utility in regional per-capita consumption (with elasticity of marginal felicity equal to 1.45), discounted with a pure rate of time preference of 1.5%. The time horizon of RICE50plus is 290 years. The maximization is solved using the GAMS software and CONOPT solver; the Nash-equilibrium is found through a fixed-point-iteration. The solution provides Lagrange multipliers for each constraint that show the value of relaxing the constraint with respect to the objective function. The regional social cost of carbon are the negative of the Lagrange-multiplier attached to the emission equation in 2025 divided by the Lagrange multiplier attached to the budget constraint in 2025 (this is similar to the social cost of carbon definition in Nordhaus (2017) but a regional instead of a global version).

The implementation of all three transfer cases and four equity principles is described in detail below.

No transfers In case of no transfers ($T_i = 0$), the BLS condition reads:

$$1 = \sum_i \frac{MB_i}{MC_i},$$

with $MB_i = scc_i$; $MC_i = mx_i[a_i \cdot x_i/z_i^B + b_i \cdot (x_i/z_i^B)^4]$. For the four equity principles, the BLS condition is solved requiring: $C_i/n_i = c_0 \forall i$ (Equal consumption), $G_i/Y_i = r_0 \forall i$ (Costs proportional to GDP), $z_i^B - x_i = z_0 \forall i$ (Equal emissions), $x_i = r_0 z_i^B \forall i$ (Equal relative provision).

Unlimited transfers In case of unlimited transfers, the BLS condition is solved while adding the constraint that carbon prices in Market Exchange Rates are equalized: $MC_i/m2p_i = a \forall i$. With parameters of RICE50plus, the carbon price is 50 (MER) 2005 USD in all scenarios, which is the global social cost of carbon in Market Exchange Rates. Regional consumption is positive with transfers. Next to these conditions, we require for the Equal consumption and Costs proportional to GDP equity principles: $C_i/n_i = c_0 \forall i$ (Equal consumption), $(G_i + T_i \cdot m2p_i)/Y_i = r_0 \forall i$ (Costs proportional to GDP). For Equal relative provision and Equal emissions, the equivalence approach is implemented. Reference abatement X_{ref} is

80% of total unregulated emissions. For Equal relative provision, utility of regions is set to: $U_i = n_i \log \left(\frac{Y_i - G_i(r_0 \cdot z_i^B) + scc_i \cdot X_{ref}/1000}{n_i} \right); \forall i$. For Equal emissions, we further allow equivalent emissions of countries to be above the unregulated level z_i^B as otherwise the scenario would be infeasible (i.e. there is no equal emission level in the equivalent utility that also solves the BLS condition). If a region's equivalent emissions are above unregulated emissions, it receives a subsidy of $\sigma=21$ USD/tCO₂ while abatement costs are zero. The subsidy is equal to globally averaged fossil fuel subsidies over the years 2017-2021.¹⁶ Utility of regions is set to: $U_i = n_i \log \left(\frac{Y_i - H(1 - z_0/z_i^B) \cdot G_i(z_i^B - z_0) + H(z_0/z_i^B - 1) \cdot \sigma \cdot (z_0 - z_i^B) \cdot n_i/1000 + scc_i \cdot X_{ref}/1000}{n_i} \right); \forall i$. Here, $H(\cdot)$ is the Heavyside function.¹⁷

Carbon market transfers With carbon markets, trade of permit endowments at individual carbon prices determines regional transfers. Total allowable emissions permits are equal to the case of unlimited transfers (denoted X_{market}). Initial permit allocation either follows a grandfathering approach, where permit endowment is proportional to each region's unregulated total emissions: $x_i^* = \frac{n_i z_i^B}{\sum_j n_j z_j^B} X_{market}$, or follows an equal-per-capita basis: $x_i^* = z_i^B - \frac{\sum_j n_j z_j^B - X_{market}}{\sum_j n_j}$. Transfers are $T_i = MC_i/m2p_i \cdot (x_i - x_i^*) \cdot \frac{n_i}{1000}$. Market clearing requires $\sum_i n_i x_i = \sum_i n_i x_i^*$. The four equity principles are implemented based on the equivalence approach, where values between regions may vary and an inequality-averse objective function is optimized. For the Equal consumption principle, utility of regions is $U_i = n_i \log \left(\frac{n_i \cdot v_i + scc_i \cdot X_{ref}/1000}{n_i} \right)$. For the Costs proportional to GDP principle, utility of regions is $U_i = n_i \log \left(\frac{Y_i - v_i \cdot Y_i + scc_i \cdot X_{ref}/1000}{n_i} \right)$. For the Equal relative provision and Equal emissions principles, utilities of regions are equal to $U_i = n_i \log \left(\frac{Y_i - H(w_i) \cdot G_i(w_i) + H(-w_i) \cdot \sigma \cdot (-w_i \cdot n_i) + scc_i \cdot X_{ref}/1000}{n_i} \right)$ with $w_i = v_i \cdot z_i^B \forall i$ (Equal relative provision) and $w_i = z_i^B - v_i \forall i$ (Equal emissions). Reference abatement is again 80% of total unregulated emissions. The objective function is $\sum_i n_i (v_i)^{(1-\eta)}/(1-\eta)$ for Equal consumption and Equal emissions, and $\sum_i n_i (1 - v_i)^{(1-\eta)}/(1-\eta)$ for Costs proportional to GDP and Equal relative provision. Inequality aversion is $\eta = 3$.

The numerical model is solved with the GAMS software using the CONOPT solver.

Solutions have been verified by maximizing the social welfare function $\sum_i \alpha_i n_i U_i$ with scenario-specific weights. For the case of no and unlimited transfers, the weights are $\alpha_i n_i \propto (C_i + B_i)/MC_i$, where variables are at the levels of the solution. For carbon market transfers the weights $n_i \alpha_i$ are set to the shadow price attached to the equivalent utility constraint.

¹⁶Data is the indicator ER.FFS_CMPT_GDP from UN statistics <https://unstats.un.org/sdgs/dataportal/database>, downloaded on Oct 10 2023.

¹⁷Numerically, we approximate the Heavyside function with the errorfunction.

Appendix D: Welfare weights

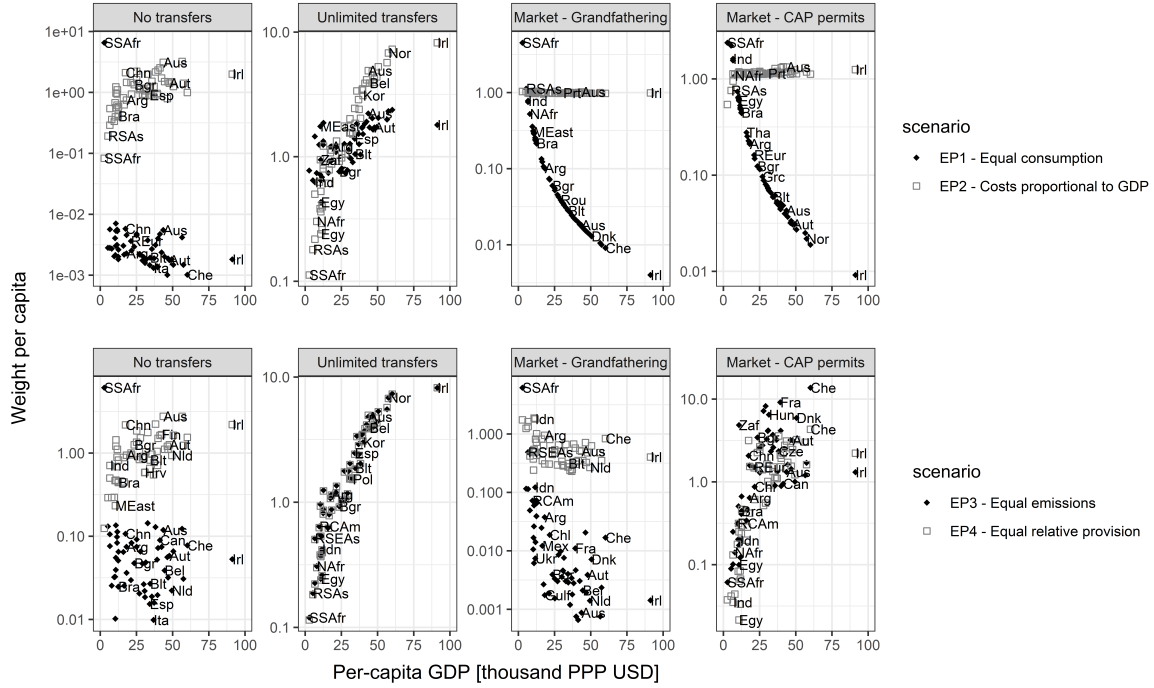


Figure 6: Weights per capita in the social optimum against GDP in Purchasing Power Parity for different scenarios. Panels show the transfer cases: no transfers, unlimited transfers, and carbon market transfers with initial permits either grandfathered or based on equal-per-capita-emissions. The transfer cases are solved for four equity principles: Equal consumption, Costs proportional to GDP, Equal emissions and Equal relative provision.

Figure 6 reports the welfare weights for all scenarios. Equal weights per person correspond to a value of 1, which have been frequently implemented in the literature (Dennig et al., 2015; Adler et al., 2017; Anthoff and Emmerling, 2019; Kornek et al., 2021). Considering the principles Costs proportional to GDP and Equal relative provision in Figure 6, weights are close to 1 (except for unlimited transfers) and increase slightly with income irrespective of the transfer scenario. Equal weights of 1 per person emerge, roughly, when the equity principle Equal consumption is implemented with unlimited transfers: with these weights, equalizing consumption through transfers is optimal. Weights decrease significantly with income when carbon prices have to be used to implement a significantly lower burden for lower income regions. This is the case for Equal consumption under no transfers and in both carbon market scenarios. It also holds true for Equal emissions unless significant transfers are available to decrease the burden for lower income regions, which is the case for unlimited transfers and a carbon market with equal per capita endowments with permits.

References

- Adler, M., D. Anthoff, V. Bosetti, G. Garner, K. Keller, and N. Treich (2017). Priority for the worse-off and the social cost of carbon. *Nature Climate Change* 7(6), 443–449.
- Adler, M., D. Anthoff, V. Bosetti, G. Garner, and N. Treich (2017). Priority for the worse off and the social cost of carbon. *Nature Climate Change* 7, 443–449.
- Adler, M. and N. Treich (2015). Prioritarianism and climate change. *Environmental and Resource Economics* 62, 279–308.
- Anke, C.-P., H. Hobbie, S. Schreiber, and D. Möst (2020). Coal phase-outs and carbon prices: Interactions between eu emission trading and national carbon mitigation policies. *Energy Policy* 144, 111647.
- Anthoff, D. and J. Emmerling (2019). Inequality and the social cost of carbon. *Journal of the Association of Environmental and Resource Economists* 6(2), 243–273.
- Bataille, C., C. Guivarch, S. Hallegatte, J. Rogelj, and H. Waisman (2018). Carbon prices across countries. *Nature Climate Change* 8(8), 648–650.
- Bauer, N., C. Bertram, A. Schultes, D. Klein, G. Luderer, E. Kriegler, A. Popp, and O. Edenhofer (2020). Quantification of an efficiencyâsoverignty trade-off in climate policy. *Nature* 588, 261–266.
- Blanchard, O., C. Gollier, and J. Tirole (2022). The portfolio of economic policies needed to fight climate change. *Annual Review of Economics* 15.
- Bode, S. (2004). Equal emissions per capita over time—a proposal to combine responsibility and equity of rights for post-2012 ghg emission entitlement allocation. *European Environment* 14(5), 300–316.
- Bretschger, L. (2013). Climate policy and equity principles : Fair burden sharing in a dynamic world. *Environment and Development Economics* 18, 517–536.
- Budolfson, M., F. Dennig, M. Fleurbaey, A. Siebert, and R. Socolow (2017). The comparative importance for optimal climate policy of discounting, inequality and catastrophes. *Climatic Change* 145, 481–494.
- Chichilnisky, G. and G. Heal (1994). Who should abate carbon emissions?: An international viewpoint. *Economics Letters* 44(4), 443–449.
- Chichilnisky, G. and G. Heal (2000). *Environmental markets: Equity and efficiency*. Columbia University Press.

- Chichilnisky, G., G. Heal, and D. Starrett (2000). Equity and efficiency in environmental markets: global trade in co2 emissions. *Environmental Markets: Equity and Efficiency*.
- Cramton, Peter, O. A. and S. Stoft (2015). An international carbon-price commitment promotes cooperation. *Economics of Energy & Environmental Policy* 4(2), 51–64.
- de Villemeur, E. B. and J. Leroux (2011). Sharing the cost of global warming. *The Scandinavian Journal of Economics* 113(4), 758–783.
- Dennig, F., M. B. Budolfson, M. Fleurbaey, A. Siebert, and R. H. Socolow (2015). Inequality, climate impacts on the future poor, and carbon prices. *Proceedings of the National Academy of Sciences* 112(52), 15827–15832.
- Flachsland, C., R. Marschinski, and O. Edenhofer (2009). To link or not to link: benefits and disadvantages of linking cap-and-trade systems. *Climate Policy* 9(4), 358–372.
- Fleurbaey, M. and U. Kornek (2021). When redistribution makes personalized pricing of externalities useless. *Journal of Public Economic Theory* 23(2), 363–375.
- Gauthier, S. and G. Laroque (2009). Separability and public finance. *Journal of Public Economics* 93(11-12), 1168–1174.
- Gazzotti, P., J. Emmerling, G. Marangoni, A. Castelletti, K.-I. v. d. Wijst, A. Hof, and M. Tavoni (2021). Persistent inequality in economically optimal climate policies. *Nature Communications* 12(1), 3421.
- Heal, G. (2000). Emissions constraints, emission permits, and marginal abatement costs, 1 in Graciela Chichilnisky and Geoffrey Heal, eds., *Environmental Markets*.
- Heal, G. and Y. Lin (2000). Chapter 5. equilibrium and efficiency: International emission permits markets. In *Environmental Markets: Equity and Efficiency*, pp. 82–109. Columbia University Press.
- Hirst, D. and M. Keep (2018). Carbon price floor (cpf) and the price support mechanism. *House of Commons Library Briefing Paper* 20.
- Jacobs, B. and R. A. De Mooij (2015). Pigou meets mirrlees: On the irrelevance of tax distortions for the second-best pigouvian tax. *Journal of Environmental Economics and Management* 71, 90–108.
- Kornek, U., D. Klenert, O. Edenhofer, and M. Fleurbaey (2021). The social cost of carbon and inequality: When local redistribution shapes global carbon prices. *Journal of Environmental Economics and Management* 107, 102450.

- Lange, A., C. Vogt, and A. Ziegler (2007). On the importance of equity in international climate policy: An empirical analysis. *Energy Economics* 29(3), 545–562.
- Leroutier, M. (2022). Carbon pricing and power sector decarbonization: Evidence from the uk. *Journal of Environmental Economics and Management* 111, 102580.
- Levi, S., C. Flachsland, and M. Jakob (2020). Political economy determinants of carbon pricing. *Global Environmental Politics* 20(2), 128–156.
- Maniquet, F. and Y. Sprumont (2004). Fair production and allocation of an excludable nonrival good. *Econometrica* 72(2), 627–640.
- Mattoo, A. and A. Subramanian (2010). Equity in climate change: an analytical review. *World Bank Policy Research Working Paper* (5383).
- Moulin, H. (1987). Egalitarian-equivalent cost sharing of a public good. *Econometrica: Journal of the Econometric Society*, 963–976.
- Nordhaus, W. D. (2017). Revisiting the social cost of carbon. *Proceedings of the National Academy of Sciences* 114(7), 1518–1523.
- Riahi, K., D. P. Van Vuuren, E. Kriegler, J. Edmonds, B. C. O’Neill, S. Fujimori, N. Bauer, K. Calvin, R. Dellink, O. Fricko, et al. (2017). The shared socioeconomic pathways and their energy, land use, and greenhouse gas emissions implications: An overview. *Global environmental change* 42, 153–168.
- Sandmo, A. (2007). Global public economics: Public goods and externalities. *Économie publique/Public economics* (18-19).
- Sheeran, K. A. (2006). Who should abate carbon emissions? a note. *Environmental and Resource Economics* 35, 89–98.
- Shiell, L. (2003). Equity and efficiency in international markets for pollution permits. *Journal of Environmental Economics and Management* 46(1), 38–51.
- Stiglitz, J. E. (2019). Addressing climate change through price and non-price interventions. *European Economic Review* 119, 594–612.
- Van Der Wijst, K.-i., F. Bosello, S. Dasgupta, L. Drouet, J. Emmerling, A. Hof, M. Leimbach, R. Parrado, F. Piontek, G. Standardi, et al. (2023). New damage curves and multimodel analysis suggest lower optimal temperature. *Nature Climate Change* 13(5), 434–441.
- World Bank (2023). State and trends of carbon pricing 2023. <http://hdl.handle.net/10986/39796>.